

Joint Master in Global Economic Governance and Public Affairs



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This work bridges my deep-rooted interest in the political economy of development with my growing commitment to social change through behavioural interventions. It represents not only what I have been focused on, but also the passions I am beginning to embrace. This thesis marks the beginning of an exciting journey towards finding my place in the world, and I am eager to see where it leads.

Abstract

This thesis examines the impact of education on improving women's conditions in China and India, focusing on the disparity between the two countries' rapid economic growth and the persistent gender inequalities in the labour market. Despite significant advancements in GDP, both nations continue to experience substantial gender discrimination in employment and wages. Operating within a multidisciplinary approach that combines development economics with behavioural economics, this research investigates the effectiveness of education as a tool to bridge gender gaps. The methodology includes a comparative analysis of educational policies and labour market trends in China and India, supported by both quantitative data and qualitative case studies. The findings reveal that while education has positively influenced female socio-economic status, structural biases and societal norms continue to impede progress and access to the workforce. To address these challenges, the thesis proposes the establishment of localized nudge units designed to implement behavioural interventions aimed at reducing gender bias and promoting equality in the labour market. This innovative approach offers a practical framework for policymakers to enhance the efficacy of educational initiatives and support sustainable economic development.

Keywords: Gender, Education, Development Economics, Labour Market, Educational Policies, China, India, Sustainable Development, Behavioural Economics, Psychology, Inequalities, Gender Biases, Economic Growth, Nudging, Female Empowerment, Behavioural Policies

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Introduction

As stated by Pareto in 1906 (2014, p.20):

“The foundation of political economy and, in general, every social science, is evidently psychology. A day may come when we shall be able to deduce the laws of social science from the principles of psychology [...].”

The relevance and importance of this study are grounded in Pareto’s assertion and the growing contemporary need to understand the psychological foundations of social behaviours and policies. This is particularly crucial in the context of gender and education within the framework of development economics. This thesis examines educational programs and policies targeted at women and girls in China and India from the early 1970s to 2022, focusing on their implications for gender inequality and the labour market. The core objective is to determine whether educational and demographic interventions contribute to gender equality and sustainable development or merely correlate with economic growth. This distinction is crucial, as economic growth alone has not ensured equitable or sustainable progress (United Nations Commission for Social Development, 2013).

Indeed, over recent decades, China and India have experienced significant growth in their economic development. However, such advancements have not been evenly distributed, as evidenced by rising Gini indexes and persistent gender inequalities. This research explores how educational policies can and did mitigate such disparities, highlighting the need for a behavioural shift to ensure gender equality beyond education, namely in an efficient allocation of resources and participation of genders in society. To address these objectives, the research is guided by the following questions.

- How have educational policies impacted gender parity in education?
- In which ways has economic growth and development influenced FLFP? How do educational attainments translate into labour market outcomes in these countries?
- How have educational policies reshaped societal norms and gender roles in China and India? What cultural and regional barriers continue to impede gender equality despite policy interventions?
- What opportunities exist for improving these policies to better support gender equality and inclusive sustainable growth?

The theoretical framework of this study is grounded in human capital theory, as articulated by Schultz (2002), which posits that education is an investment that needs to be directed to women, to enhance skills and productive knowledge, thereby increasing economic productivity. Yet, policies must address present biases among parents, who may undervalue the long-term benefits of educating daughters. The present research will not cover other forms of gender inequality unrelated to education and access to the labour market or explore in-depth comparisons with countries outside of China and India.

The thesis is structured as follows. The Literature Review examines existing research on educational policies, gender equality, and economic development. The Methodology chapter outlines the research design, data collection, and analysis methods. Chapters I (“Do Chinese Women Truly Hold Up Half the Sky?”) and II (“Is India’s Temperature Rising or Falling?”) explore the impact of China’s and India’s educational policies on gender equality and economic growth. They discuss key initiatives, outcomes, and analyse trends in inequalities, development, reproduction, educational attainment, and FLFP. The titles of these first two chapters draw inspiration from two profound statements and are used as a frame for the dissertation’s goals. The phrase “Women hold up half the sky” is attributed to Mao Zedong¹ as part of a broader strategy to mobilize the entire population for the socialist transformation. Through the emphasis placed on women’s crucial contributions, Mao aimed to break down traditional gender barriers and fully integrate them into the social and economic fabric of the nation. In a similar vein, the wisdom of Swami Vivekananda, a prominent Indian philosopher, mirrors such perspective stating that “The best thermometer to the progress of a nation is its treatment of its women” (Shetty, 2024). These provocative chapter titles set the stage for a critical examination of gender equality and sustainable development in China and India. Chapter III provides a comparative analysis, examining institutional frameworks, progress towards SDGs, and the necessity and relevance of behavioural policies and “nudge

¹ The phrase “Women hold up half the sky” was popularized during the early years of the People’s Republic of China. Mao used this slogan to emphasize the crucial role of women in the revolutionary struggle and socialist construction. In the 1950s and 1960s, the Chinese Communist Party implemented policies to promote gender equality, such as the Marriage Law of 1950, which aimed to abolish feudal practices and improve women’s rights in marriage and family life. This slogan encapsulated the CCP’s commitment to women’s liberation and their active participation in the workforce and political spheres. For a detailed analysis, see Croll, E. (1983). *Chinese Women Since Mao*. London: Zed Books.

units².” The Conclusion discusses the findings, implications for policy, and offers recommendations for promoting inclusive and sustainable development.

The relevance of this study is twofold. Academically, it contributes to the ongoing discourse on the role of education in achieving gender equality and economic development. Practically, it provides policymakers with evidence-based insights that can enhance the effectiveness of policy interventions. This is particularly pertinent for developing nations where gender disparities in access and economic inequalities remain significant challenges.

² A nudge unit refers to a team or group within an organization, often within government or public policy bodies, that employs behavioural science principles to design interventions. The concept is based on the idea of “nudging,” as popularized by Richard H. Thaler and Cass R. Sunstein in their book *Nudge: Improving Decisions About Health, Wealth, and Happiness*. These units apply insights from psychology and economics to encourage better decision-making and policy outcomes by subtly guiding individuals’ choices, often through changes in the way options are presented (Thaler and Sunstein, 2008). The concept of nudge units draws inspiration from the success of The Behavioural Insights Team, a UK-based global social purpose organisation that informs policy and improves public services, following nudge theory (The Behavioural Insights Team, 2022).

Literature Review

The interplay between education, gender equality, and the labour market is a critical area of research, particularly in rapidly developing and densely inhabited countries like China and India, whose residents account for around one third of the global population. A significant body of foundational knowledge is encapsulated in seminal books and monographs. Acemoglu's "Introduction to Modern Economic Growth" (2009) offers several insights into economic development theories, while Becker's "Human Capital" (1993) explores the critical role of education in fostering economic growth. Drèze and Sen's "An Uncertain Glory" (2013) provides an in-depth analysis of social and economic issues in India, stressing the importance of education and addressing gender inequality. Additionally, a substantial portion of the literature is found in peer-reviewed journals. For instance, Asadullah and Yalonetzky (2012) highlight significant disparities across Indian states through an empirical analysis of educational inequality. Similarly, Gustafsson and Li (2000) examine the gender earnings gap in urban China and the economic transformation's impact on gender equality. However, these studies often lack an integrated examination of how education, sustainable development, and female labour force participation interconnect specifically in relation to gender equality.

Empirical studies provide robust, data-driven insights. Duflo (2012) explores the relationship between women's empowerment and economic development, using extensive data to argue that empowering women leads to broader economic benefits. This study reviews a range of countries, including India, China, South Africa, and Mexico. Iwasaki and Ma (2020) conduct a meta-analysis on the gender wage gap in China, synthesizing findings from numerous studies to present a comprehensive picture of wage disparities. Chakraborty and Krishnankutty (2012) highlight that education boosts economic productivity and growth in India. Jariwala (2017) uses econometric methods to demonstrate how educational investments drive economic growth in India. Additionally, Marquez-Ramos and Mourelle (2019) found nonlinear relationships between education and economic growth in Spain, indicating varying impacts at different levels of educational attainment. Policy reviews and case studies offer practical insights into the implementation and impact of various policies. Aithal and Aithal's (2020) review of India's NEP 2020 assesses its objectives and potential impact on achieving educational

equity. Bhardwaj (1992) provides a detailed case study on China's SEZs, examining their role in economic reform, development, and female labour participation.

Given the rapidly evolving socio-economic landscapes in China and India, the temporal relevance of the literature is crucial. Recent publications, including ministerial documents and reports from international organizations such as the UN, UNESCO, the World Bank, and the IMF, provide valuable current insights. However, historical context remains important, with older works like Sen's (1990) study on missing women providing foundational insights that continue to influence contemporary research (Bongaarts and Guilmoto, 2015).

Research from other regions provides a broader context and valuable comparisons. In the United States, Goldin's (2014) "A Grand Gender Convergence: Its Last Chapter" explores gender roles and barriers to achieving gender equality in the labour market, underlining the necessity of changes in workplace norms and policies, such as more flexible work arrangements and better support for work-life balance. Blau and Kahn's (2017) paper, "The Gender Wage Gap: Extent, Trends, and Explanations," provides a comprehensive review of the gender wage gap in the US, analysing factors such as education, experience, occupation, and industry. Moreover, research by Kleven, Landais, and Sogaard (2018), "Children and Gender Inequality: Evidence from Denmark," offers relevant insights into the impact of children on gender inequality in the labour market. The study shows that motherhood significantly impacts women's earnings and career progression, highlighting the need for policies that support parents, such as paid parental leave and affordable childcare. In Brazil, Soares, Osório, and Ribas (2010) analyse the impact of conditional cash transfer programs like Bolsa Família, directed towards low-income families, with requirements such as school attendance for children. Their research finds that the program has significantly improved educational outcomes and reduced poverty, financially empowering women and increasing girls' school attendance. Steele and Aronson's (1995) study on stereotype threat highlights the impact of social psychology on the intellectual performance of African Americans, offering insights into how educational outcomes can be influenced by social factors and biases. These papers underline significant aspects of inequalities but often treat education, gender equality, economic growth, biases, and labour market participation as separate entities rather than interconnected factors, further emphasizing the need for integrated research.

Badel, Fabrizio, and Goyal (2024) suggest that global gender gaps may never close on their own, noting that while solutions are often proposed, they rarely address the root causes comprehensively. Their observation underscores the necessity for innovative approaches and identifies critical gaps in literature and research. Specifically, there is a lack of integrated studies that combine research on education, sustainable development, and female access to the labour force. The intersection of development and behavioural economics in the context of gender equality is also underexplored. This includes how educational attainment and economic growth affect female workforce participation and the role of behavioural policies in challenging cultural barriers and biases. One promising area for addressing these gaps in the literature and research is through the application of behavioural policies. Bicchieri's "The Grammar of Society" (2006) examines how social norms influence gender roles and labour participation. Sunstein and Thaler's (2003) concept of libertarian paternalism provides a framework for designing policies that nudge individuals towards better decision-making without restricting their freedom. Thaler (2015) discusses the principles of behavioural economics, crucial for understanding how economic policies can influence gender equality. Bohnet (2018) explores gender equality by design, arguing that thoughtfully designed policies can promote gender equality in the workplace and beyond. Addressing these gaps is essential for achieving gender parity and meeting SDG targets, making it an important area for current research and future studies.

Methodology

The methodological framework used throughout this thesis employs multidisciplinary, qualitative, and comparative approaches. The qualitative analysis enables an in-depth exploration of contextual factors influencing policies' implementation and outcomes, while the comparative analysis identifies similarities and differences between the education systems, labour market, and gender equality initiatives in the two countries. To address the subjectivity inherent in qualitative research, triangulation and validation of data from multiple sources are utilized, focusing on trends rather than one-size-fits-all results. Cross-national comparisons are carefully managed to account for contextual differences. A multidisciplinary approach integrates insights from behavioural and development economics. Behavioural economics provides a framework for understanding how psychological factors and cognitive biases influence decision-making, impacting policies and socio-economic change. Development economics offers a broader perspective on the economic, social, and institutional factors driving development and influencing gender equality.

This research employs primary and secondary data sources, along with case studies, to examine education policies and their impact on the labour market and gender equality in China and India. Primary data sources include reports, policy documents, and data from international organizations such as the UN, UNESCO, World Bank, and IMF. Secondary data sources consist of existing literature, including scholarly articles, books, and academic journals, offering insights into the historical context, policy frameworks, and implementation strategies of initiatives in both countries. Additionally, case studies on specific education policies and gender equality initiatives supplement empirical evidence. The analysis employs descriptive statistics to examine quantitative data, exploring trends, patterns, and correlations to evaluate the effectiveness of policies and programs. Visualizations such as charts and graphs dynamically represent complex datasets, aiding in the identification of patterns and trends.

Chapter I: Do Chinese Women Truly Hold Up Half the Sky?

China's Educational Policies and Initiatives Towards Gender Equality

Educational policies profoundly influence and shape societal norms and gender roles, serving as powerful indicators of the promotion or inhibition of gender equality. This chapter examines key Chinese programs impacting girls' and women's educational attainment, including the Modernization of China's Education by 2035, the Compulsory Education Law, the Spring Bud Project, and the One Child Policy.

Wu's (2022) analysis of China's changes in educational equality provides important new light on how gender equality and economic reform interact. A planned economy transitioned to a marketized one with the economic reform of 1978³, and this, alongside industrialization, created for skilled female workers unprecedented opportunities. Consequently, numerous young rural women migrated to urban areas for industrial employment, achieving financial independence (Wu, 2022). Industrialization precipitated significant structural transformations, notably rural-to-urban migration and overall development (Nayyar, 2019).

The OCP and CEL played decisive roles in this socioeconomic shift. The former fostered a more egalitarian approach to education and upbringing, altering parental attitudes toward their children (Lee, 2012). With families limited to one child, daughters received increased educational investment and personal development opportunities (Huang, Lei and Sun, 2015). Concurrently, the CEL enforced nine years of compulsory education, significantly mitigating educational inequalities and ensuring more equitable educational attainment between genders (Guo, Huang and Zhang, 2019). As Bicchieri (2006) suggests, social norms can be transformed by changing individuals' expectations about others' behaviour. In China, policies and initiatives have partially succeeded in this regard, establishing new standards for what is considered normal and acceptable concerning gender roles in education. Although significant educational improvements

³ The economic reforms of 1978 in China, initiated by Deng Xiaoping, marked a significant shift from a centrally planned economy to a market-oriented economy. These reforms introduced measures such as the de-collectivization of agriculture, the establishment of Special Economic Zones (SEZs) to attract foreign investment, the decentralization of economic decision-making, and the liberalization of prices. The reforms aimed to increase productivity and economic growth by integrating China into the global economy and encouraging private enterprise (Naughton, 2007).

have been achieved, especially in urban areas, disparities persist between urban and rural regions, particularly for girls (Zhang, Li and Xue, 2015). This chapter provides essential contextual information for the subsequent examination of economic and social trends and outcomes.

The One Child Policy

The Chinese government enforced the OCP from 1979 to 2015⁴ as a means of population control, seeing rapid population increase as a problem for resource management and economic development. A range of strategies were applied to implement the policy, including fines for noncompliance, financial incentives for compliance, and, in certain cases, coercive measures including forced sterilisations and abortions (Bongaarts and Greenhalgh, 1985). Higher fines under the OCP in adolescence were associated with a higher chance of female senior high school completion (Huang, Lei and Sun, 2015).

An estimated 400 million births have been avoided because of the OCP (Hvistendahl, 2017), greatly reducing the rate of population growth. The policy was criticised for its negative social effects, such as an ageing population and a skewed gender ratio brought on by a cultural preference for male children, which led to sex-selective abortions and female infanticide (Bongaarts and Guilmoto, 2015), but it also unintentionally made a small but significant contribution to the path towards the improvement of gender parity. The policy specifically resulted in the reallocation of family resources, such as money for education, to support single daughters, and girls' access to school significantly improved with time. Gender parity had mostly been reached in basic and secondary education by the early 21st century, and more women were already pursuing higher education. Huang, Lei and Sun's analysis (2015) suggests that OCP accounts for half of China's gender gap narrowing and thirty percent of the educational growth for women born between 1945 and 1980. From their findings, the policy fines

⁴ After 2015, the policy was replaced by the Two Child Policy, which allowed families to have two children in response to the demographic challenges posed by an aging population and shrinking workforce. However, as these measures proved insufficient to significantly increase birth rates, the government further adjusted its policy in 2021 to allow families to have up to three children (Goldman, 2021). This latest policy shift is part of broader efforts to mitigate the effects of an aging population and to sustain economic growth by encouraging higher birth rates.

were linked to several societal changes, including shifting visions on gender equality and children, delayed marriage, higher labour force participation, and altered fertility rates. However, the shrinking fertility rate has resulted in a labour shortage and more strain on the younger generation to care for the elderly, creating new socioeconomic difficulties (Settles et al., 2012).

The Compulsory Education Law

China's educational environment underwent a remarkable shift in April 1986 with the implementation of the CEL. It was the first law in the country to require compulsory education, and it included three important provisions (Ma, 2024). The first was that all children are mandated free and compulsory school for nine years, encompassing pre-primary, primary, secondary, and higher education (ibid.). The law also covered special needs education, literacy programs, open and online learning, adult education, and continuing education (Guo, Huang and Zhang, 2019). Additionally, it instructed that children start their mandatory education in September of the year they turn six (Guo, Huang and Zhang, 2019; Ma, 2024). Finally, in the year of its introduction, students under the age of 15 who had previously dropped out of school were compelled to return and finish their nine years of education, while those above 15 were not required to continue their education (Fang et al., 2012). The central government gave each province the authority to choose the effective dates of the CEL because of the existing differences in the allocation of economic and educational resources among them (Ma, 2024). As a result, the majority of provinces enforced the law between 1985 and 1991, having an impact on the subsequent children's cohorts (Du, Xiao and Zhao, 2020).

Most importantly, substantial benefits in promoting equitable decision-making processes among families and intrahousehold empowerment have been shown when exposed to the CEL. Ma's research (2024) reveals that respondents' likelihood of seeing egalitarian intrahousehold empowerment increased by 4.4 percentage points and their egalitarian intrahousehold decision-making index increased by 0.107 standard deviation units after being exposed to the CEL. Notably, the impacts were more noticeable in cities, indicating a nationwide, but regionally variable, reduction in gender gaps in intrahousehold empowerment (Ma, 2024). Higher income and more gender equality-

aligned attitudes are thus associated with better educated women, who also tend to be more empowered within their households. According to Ma's study (2024), exposure to the CEL improved the yearly income of female respondents by 11.5%, the egalitarian gender views index of all respondents climbed by 0.127 standard deviation, the age at which women first married was delayed, and the educational attainment of their spouses was higher.

The Spring Bud Project

In 1989, the China Children and Teenagers' Foundation and the All-China Women's Federation initiated the Spring Bud Project to enhance educational opportunities for girls from low-income families. This project supported girls' education and general development from pre-school to university through companionship programmes, practical technical training, adolescent education efforts, scholarships, and social practice activities, successfully decreasing the rates of dropout among girls from underprivileged families and the financial barriers to education. Between 1989 and 2013, it provided over 2.32 million scholarships and 520,000 technical training sessions, facilitating literacy eradication and the popularization of nine-year compulsory education (UNESCO International Research and Training Centre for Rural Education, 2022).

During its first phase, between 1989 and 2000, the two main goals were to end illiteracy and make compulsory schooling nine years of age for everyone. Sixteen Spring Bud girls' courses were built between 1990 and 1992 in Guangxi Zhuang autonomous regions and Guangdong, Hunan, and Sichuan provinces, enabling the return of girl dropouts to school (UNESCO International Research and Training Centre for Rural Education, 2022). Teenage girls also benefited from the establishment in 1996 of a specific fund for hands-on technological training. The second phase, from 2001 to 2011, concentrated on building Spring Bud Schools, adopting technical training programmes, and bolstering education in underprivileged and rural areas (ibid.). At this phase, the improvement of education in the nation's poorest regions was given priority. With vocational training programmes, the Spring Bud Project helped girls from low-income communities finish their education and improve their chances of finding work. Since 2012, the project has focused on providing equal and high-quality education, tackling

gender-based violence and mental health issues (Women of China, 2023). Programs like the Spring Bud Companionship Programme and adolescent education initiatives furthered female empowerment, coupled with summer camps and teacher training initiatives which also enhanced educational opportunities (UNESCO International Research and Training Centre for Rural Education, 2022). In 2023, the Spring Bud Project received the UNESCO Prize for Girls' and Women's Education (UNESCO, 2024). Through the garnering of broad social support, the project's long-term influence has been to disrupt traditional gender norms, elevating females' status in households and society, encouraging girls to participate in public welfare, advancing social and sustainable development within communities. The project has developed a trained female workforce through education and vocational training, promoting well-being by increasing knowledge of health, hygiene, self-protection, self-confidence, and self-reliance (UNESCO International Research and Training Centre for Rural Education, 2022). In addition to scientific literacy and other 21st-century competences, girls acquire social and emotional skills that equip them for meaningful engagement in society and the workforce.

Modernization of China's Education by 2035

The objective of China's education modernization by 2035 is to cultivate a well-educated and skilled workforce that supports a prosperous, strong, and balanced nation (Ministry of Education of the People's Republic of China, 2019). Key development goals

“[...] include the establishment of a modern education system that provides lifelong learning for all, the universalization of quality preschool education, the achievement of high-quality and balanced compulsory education, the complete universalization of upper secondary education, the significant improvement of vocational education services, marked improvement in the competitiveness of higher education, the provision of the appropriate education for disabled children, and the formation of a new pattern of education management involving the participation of society” (Zhu, 2019, p.357).

Since innovation is the driving force behind education modernization, China's 2035 goals outline actions to improve educational development, including modernizing governance, utilizing new technologies, and innovating educational services (Ministry of Education of the People's Republic of China, 2019). This includes upgrading talent development, building a professional teaching force, and leveraging ICT for education reform (Zhu, 2019). Considering that by improving the quality and accessibility of

education for women there is a direct impact on their participation in the labour force, the plan could help prepare females to compete in a dynamic job market by fostering an education system that emphasizes well-rounded development and practical skills. Indeed, enhanced vocational and higher education opportunities provide women with the skills needed for high-demand industries, promoting their entry into traditionally male-dominated fields (Ministry of Education of the People's Republic of China, 2019).

Furthermore, the modernization plan advocates for a new education management system involving societal participation beyond governmental support, including community and family involvement, which is crucial for changing and challenging traditional gender norms. The plan must address the diverse needs and aspirations of the youth in a globalized context, through the creation of opportunities for international exchanges and ensuring that Chinese education remains relevant to the global job market. Effectively engaging youth will necessitate innovative approaches and a nuanced understanding of their evolving needs and perspectives, for both female and male workforces.

Chinese Women Empowerment and Transformation

China's transformation over the past few decades has been profound, marked by rapid economic growth, educational advancements, and impactful demographic shifts. Despite these steps forward, several challenges persist, particularly in terms of inequalities. This section provides a comprehensive analysis of these trends, incorporating data and graphs retrieved from Our World in Data⁵ to highlight key patterns in education, demographic changes, economic growth, human development indices, and gender equality, underpinned by relevant literature. The analysis of trends begins with examining the GDI⁶ patterns, providing a foundation for the subsequent scrutiny. In China, the GDI started at a relatively high value in 1990, reflecting progress from the

⁵ Our World in Data is an open-access, web-based publication that provides extensive research and data on some of the world's largest problems, such as poverty, disease, hunger, climate change, war, existential risks, and inequality. Founded by Max Roser, a researcher at the University of Oxford, the platform presents empirical data and research in an accessible and understandable format for a broad audience, including policymakers, researchers, educators, and the general public. It compiles data from multiple sources and presents it through interactive charts and visualizations to illustrate long-term trends and provide context for understanding global issues.

⁶ The Gender Development Index measures gender inequalities in life expectancy, education, and income.

socialist policies implemented post-1949 that emphasized female workforce participation and educational opportunities (Cao, 2022). As shown in Figure 1, there was a slight increase in the mid-1990s, corresponding to the consolidation of advances resulting from the CEL (Guo, Huang and Zhang, 2019). The GDI continued to grow steadily from 2000 to 2015, a period characterized by rapid economic development, urbanization, significant investments in education and healthcare, and regulations promoting gender equality (Tang, 2023). Concurrently, the GII decreased from around 0.3 in 1998 to approximately 0.186 in 2022 (United Nations Development Program, 2022).

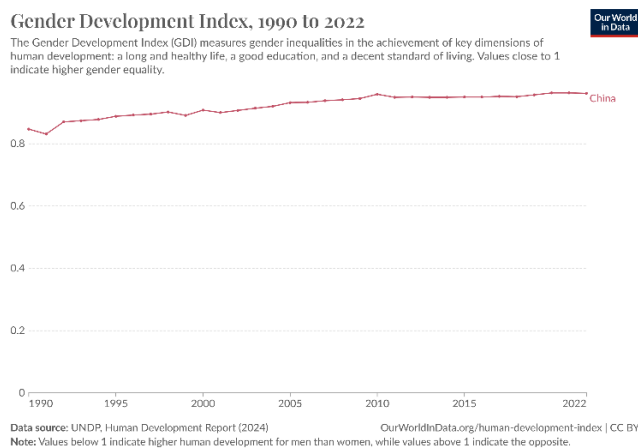


Figure 1. Gender Development Index in China (1990-2022).
Source: *Our World in Data* (2024q).

To substantiate the significant advancements illustrated in the literature, the HDI⁷ confirms the notable progress over the considered period, with an increase from approximately 0.5 in 1990 to around 0.8 in 2022 (Our World in Data, 2024t).

⁷ The Human Development Index offers a comprehensive measure of development that extends beyond economic growth, encompassing the key dimensions of life expectancy, education, and standard of living.

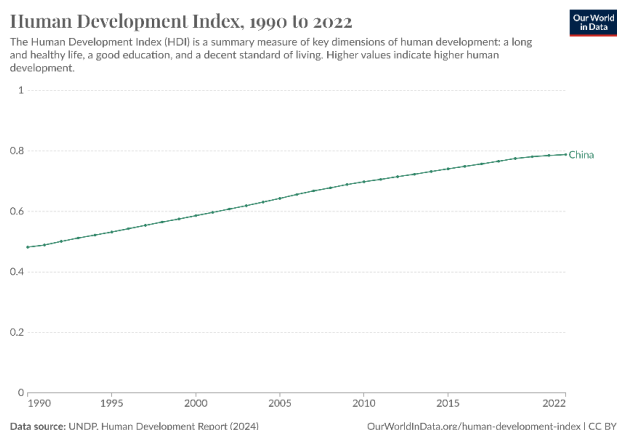


Figure 2. Human Development Index in China (1990-2022).

Source: *Our World in Data* (2024t).

China			
2022 GII value	0.186		
GI change from 2021	+0.001		
Maternal Mortality Ratio	23.0 death/100,000 live births		
Adolescent Birth Rate	11.1 births/1,000 women age 15-19		
	Female	Male	Gender gap
Share of seats in parliament	24.9%	75.1%	-50.1%
Population with at least some secondary education (age 25 and older)	79.7%	86.4%	-6.7%
Labour force participation rate (age 15 and older)	53.8%	74.5%	-20.8%

Figure 3. China's Gender Inequality Index (2022).

Source: *Human Development Reports* (UNDP, 2022).

Upon further analysis, following the establishment of the People's Republic of China, data indicate an increase in average years of schooling (Figure 4). The CPC's initiatives to instil socialist values, mobilize for revolution, and implement economic reforms led to substantial investments in education, marking significant improvements in modernization and national revitalization (Guo, Huang and Zhang, 2019). However, around 2020, Figures 4 and 5 show a plateau, suggesting the need to address deeper systemic issues such as education quality and urban-rural disparities, rooted in the hukou system⁸ (Huang et al., 2022). This system benefits urban residents, resulting in unequal resource allocation and limited mobility for rural inhabitants (Chan, 2010). Rural schools often lag behind urban ones in both quality and quantity, intensifying educational

⁸ The household registration system, or hukou, is a mechanism in China that registers individuals as either rural or urban residents. It dictates access to various social services, including education, healthcare, and housing. Rural residents often face significant restrictions and limited access to these services in urban areas, perpetuating socio-economic inequalities (Chan, 2010).

inequality. Urban-centric key schools exacerbate this issue, while mobility restrictions prevent rural migrants from accessing urban education (Tang, 2023).

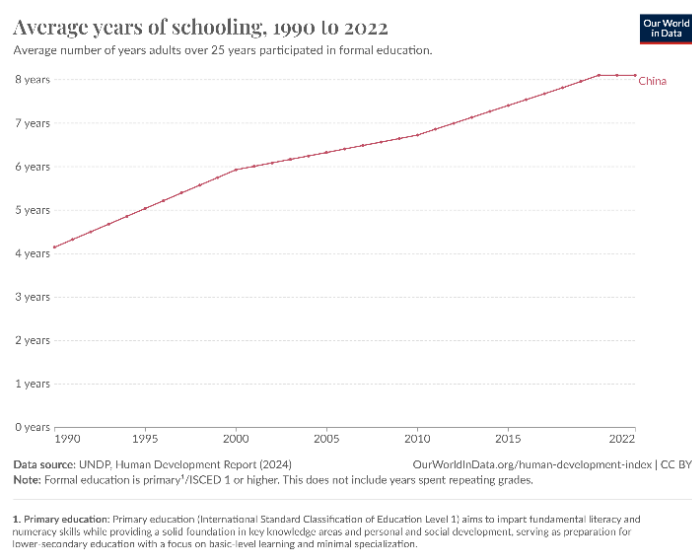


Figure 4. Average Years of Schooling in China (1990-2022).

Source: Our World in Data (2024h).

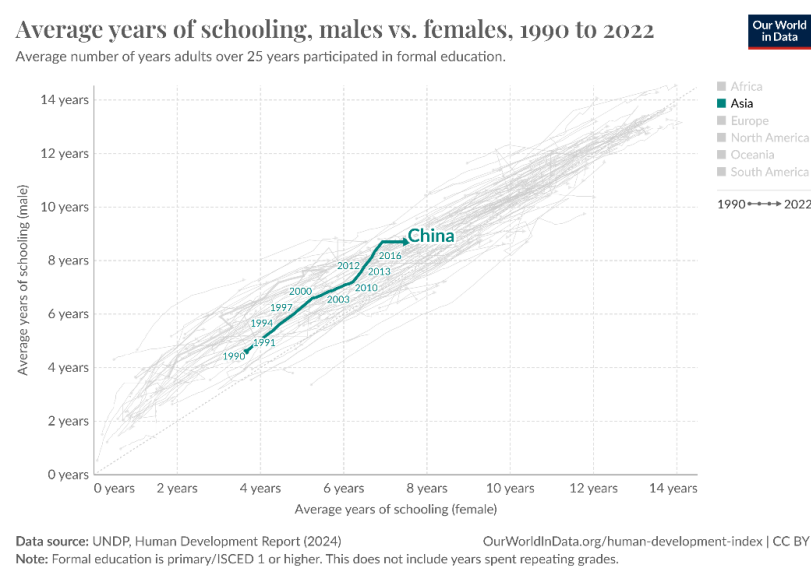


Figure 5. Average Years of Schooling by Gender in China (1990-2022).

Source: Our World in Data (2024i).

Expected years of schooling, males vs. females, 2022

The number of years a child of school entrance age can expect to receive if the current age-specific enrollment rates persist throughout the child's years of schooling.



Data source: UNDP, Human Development Report (2024)

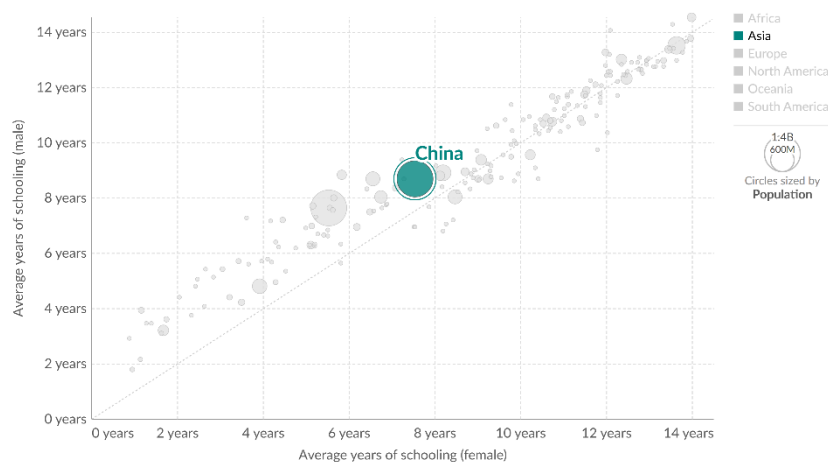
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Figure 6. Expected Years of Schooling by Gender (2022).

Source: *Our World in Data* (2024i).

Average years of schooling, males vs. females, 2022

Average number of years adults over 25 years participated in formal education.



Data source: UNDP, Human Development Report (2024)

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Note: Formal education is primary/ISCED 1 or higher. This does not include years spent repeating grades.

Figure 7. Average Years of Schooling by Gender in China (2022).

Source: *Our World in Data* (2024i).

China has achieved near-equal expected and average years of schooling for both genders, with both respectively averaging around 14-15 years and 8-9 years (Figure 6 and 7). Before 1978, educational opportunities for women were limited due to traditional Confucian values and socio-economic barriers. Only about 20% of females received any formal education, and the literacy rate among women was very low (Cao, 2022). The increase slows after 2015, plateauing at around 14 years: ensuring high-quality education, where resources and opportunities are limited, especially in rural areas, remains indeed a critical challenge (Guo, Huang and Zhang, 2019). However, rapid economic growth has enabled increased funding for education, allowing for the construction of new schools, training of teachers, and provision of educational resources (ibid.). Moreover, China's position in Figure 8 reflects overall strong educational outcomes in relation to the quantity of schooling: the size of the circle representing China indicates its large population, underlining the scale at which these outcomes are achieved. Chinese quality of education index is around 0.65, placing it relatively high but still below the top-performing countries (World Population Review, 2021). Continuous reforms aiming at aligning the system with international standards have contributed to the improvement in learning outcomes and education quality, in addition to programs focusing on STEM (Guo, Huang and Zhang, 2019). Nonetheless, rural areas continue to lag in educational resources and opportunities, reflecting the enduring impact of the hukou system (Tang, 2023).

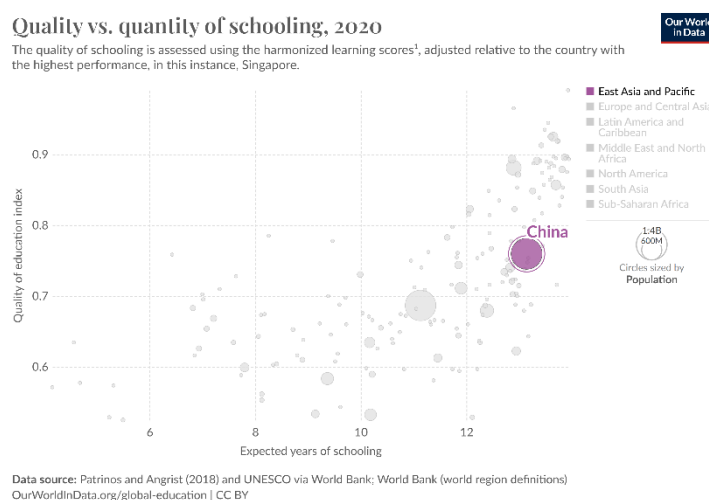


Figure 8. Quality vs. Quantity of Schooling in China (2020).

Source: *Our World in Data* (2024ab).

1. **Harmonized test scores:** Harmonized test scores consolidate data from several international student achievement testing programs, enabling a standardized comparison of educational attainment across different educational systems and cultures. These scores are measured in TIMSS (Trends in International Mathematics and Science Study) - equivalent units, with 300 denoting minimal attainment and 625 representing advanced attainment.

To study and investigate the patterns in literacy outcomes, it is essential to compare female literacy rates with those of adults and men over time. The adult literacy rate increased from approximately 65% in 1982 to over 95% by 2020 (Our World in Data, 2024a). Similarly, the female youth literacy rate rose from about 80% in 1982 to nearly 100% by 2020, demonstrating significant progress in closing gender gaps through targeted programs and policies (Cao, 2022). The rapid increase in literacy during the 1980s and 1990s was largely due to government initiatives aimed at addressing historical gender biases and barriers to education (Tang, 2023). Additionally, focusing on marginalized and rural populations was crucial in achieving near-universal literacy (ibid.) This progress in literacy rates (Figures 9-12) is mirrored by the increase in the percentage of the population with at least some formal education (Figure 13).

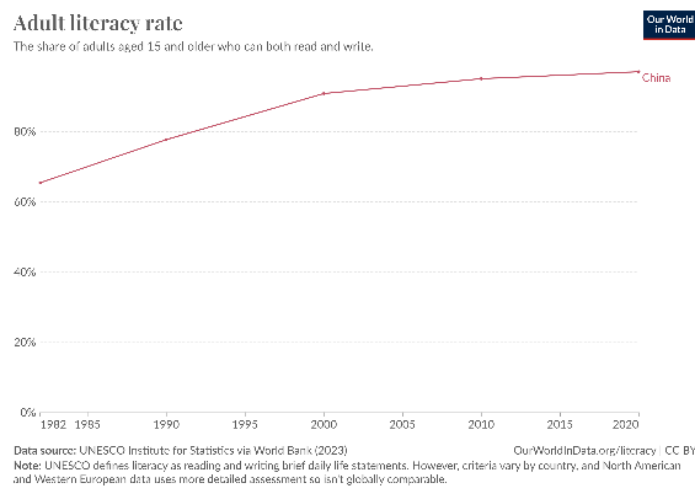


Figure 9. Adult Literacy Rate in China (1982-2020)

Source: *Our World in Data* (2024a).

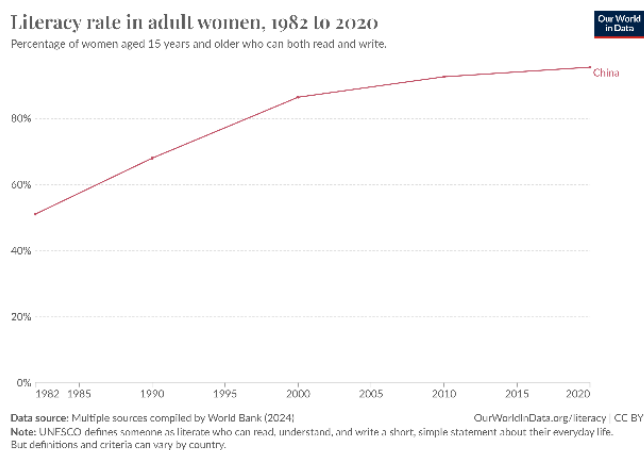


Figure 10. Literacy Rate Among Adult Women in China (1982-2020).

Source: *Our World in Data* (2024b).

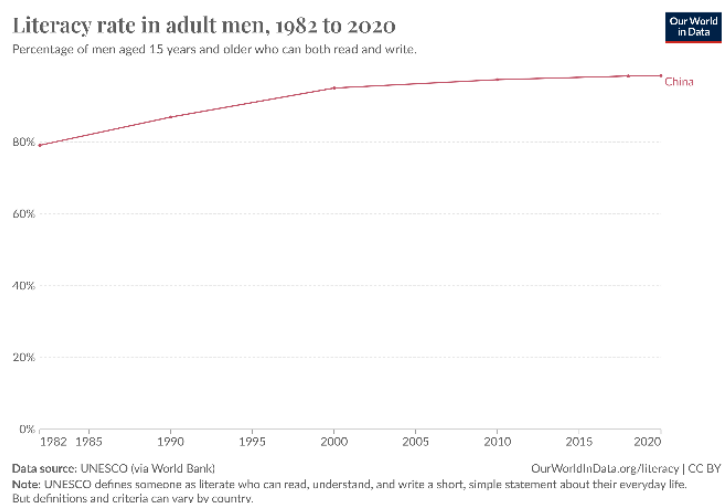


Figure 11. Literacy Rate Among Adult Men in China (1982-2020).

Source: *Our World in Data* (2024c).

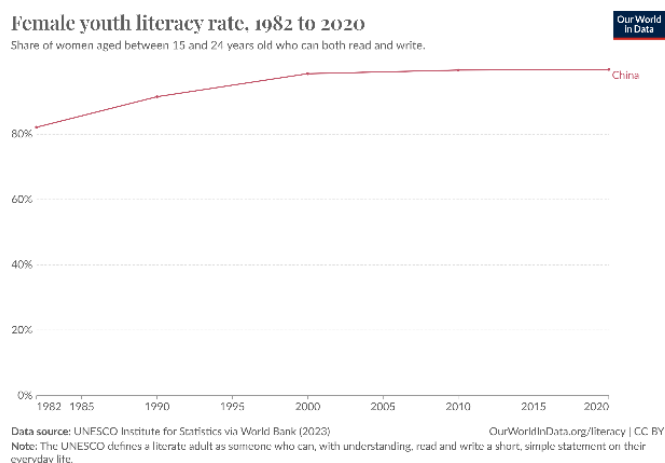


Figure 12. Female Youth Literacy Rate in China (1982-2020).

Source: *Our World in Data* (2024n).

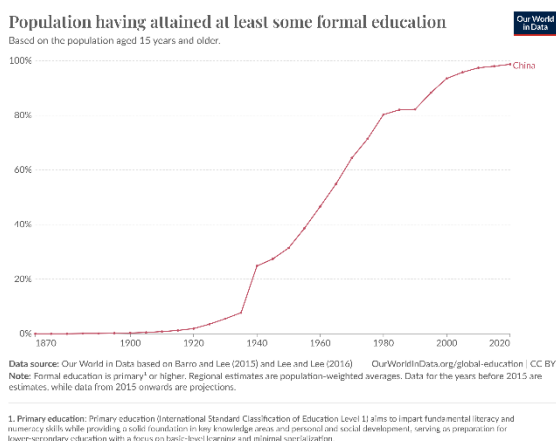


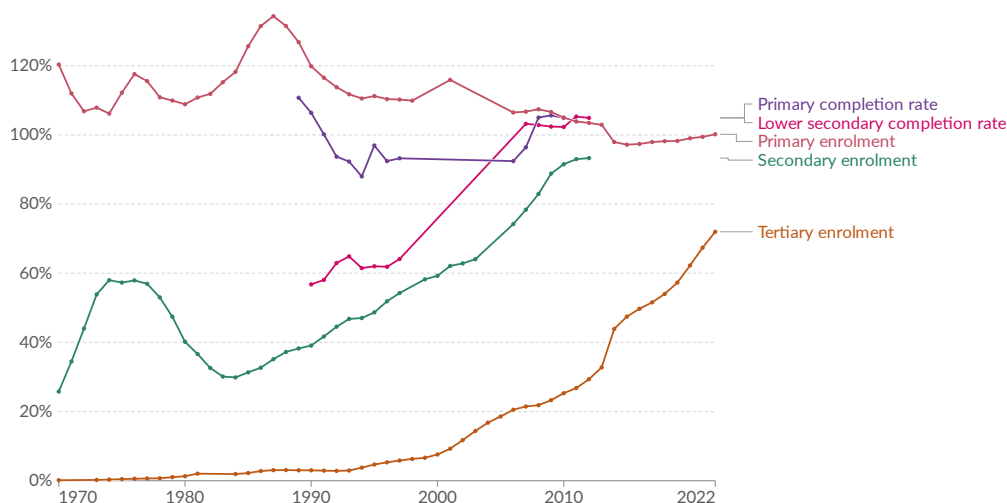
Figure 13. Population with at Least Some Formal Education in China (1870-2020).

Source: *Our World in Data* (2024x).

Primary, secondary and tertiary education enrolment and completion rates, China



This is given as the gross rate, which includes children of any age entering the level of education; this can result in percentages greater than 100 because children may enter education late or repeat a year.



Data source: World Bank, Gender Statistics (based on multiple sources); UNESCO via World Bank
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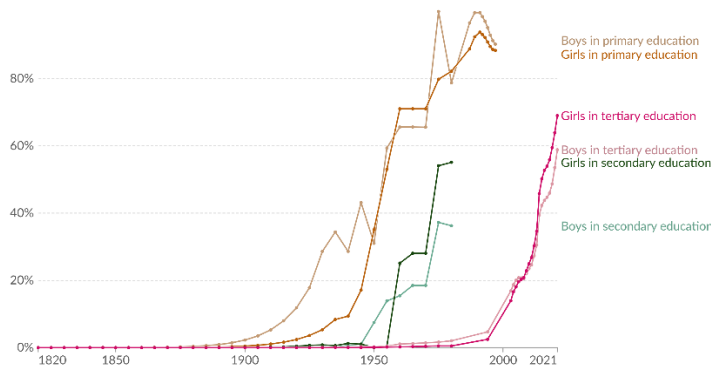
Figure 14. Enrolment and Completion Rates for Primary, Secondary, and Tertiary Education in China (1970-2022).

Source: *Our World in Data* (2024y).

Gender gap in primary, secondary and tertiary education, China

Share of boys or girls within the relevant age group who are enrolled in primary¹, secondary², and tertiary³ education.

Our World
in Data



Data source: Our World in Data based on Lee and Lee (2016) and UNESCO via World Bank

Note: This is given as the 'gross' rate, which includes children of any age entering the level of education; this can result in percentages greater than 100 because children may enter education late or repeat a year.

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- 1. Primary education:** Primary education (International Standard Classification of Education Level 1) aims to impart fundamental literacy and numeracy skills while providing a solid foundation in key knowledge areas and personal and social development, serving as preparation for lower-secondary education with a focus on basic-level learning and minimal specialization.
- 2. Secondary education:** Secondary education (International Standard Classification of Education Level 2 and 3) completes the provision of basic education that began at the primary level, and aims at laying the foundations for lifelong learning and human development, by offering more subject- or skill-oriented instruction using more specialized teachers.
- 3. Tertiary education:** Tertiary education (International Standard Classification of Education Level 5 to 8) expands upon secondary education by offering specialized learning activities in various fields. It targets advanced levels of complexity and specialization, encompassing both academic and advanced vocational or professional education.

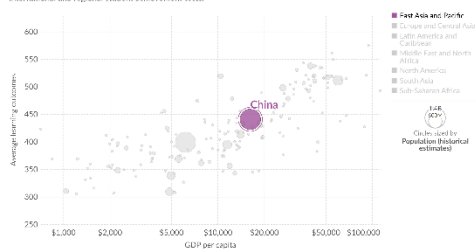
Figure 15. Gender Gap in Primary, Secondary, and Tertiary Education in China (1820-2021).

Source: Our World in Data (2024r).

Average learning outcomes vs. GDP per capita, 2020

Average learning outcomes correspond to harmonized scores across standardized, psychometrically-robust international and regional student achievement tests.

Our World
in Data



Data source: PISA and AVEL (2018) and UNESCO via World Bank. Data compiled from multiple sources by World Bank

Note: GDP per capita data is expressed in International \$ at 2017 prices.

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- 1. Harmonized test scores:** Harmonized test scores consolidate data from various international student achievement testing programs, enabling a standardized comparison of educational attainment across different educational systems and cultures. These scores are reported in PISA (Trends in International Mathematics and Science Study) - equivalent units, with 200 denoting minimal attainment and 625 representing advanced achievement.
- 2. International dollars:** International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: [What are International Dollars? How Do They Work and Why Do They Matter?](#)

Figure 16. Average Learning Outcomes vs. GDP per Capita in China (2020).

Source: Our World in Data (2024g).

Primary enrolment rate started above 100%⁹ in the 1970s, experienced some fluctuations, and has then been stable since the early 1990s, while primary completion rate generally being close to 100% from the mid-1990s onward (Figure 14). Secondary enrolment was relatively low around 40% in the early 1970s, declined in the late 1970s, then started a steady increase from the mid-1980s, reaching around 95% by 2022. Tertiary enrolment saw a dramatic boost from near 0% in the 1970s to above 50% by 2022, highlighting Chinese expansion and increased accessibility of higher education. To analyse gender parity in education, it is important to examine the gender gap throughout the schooling years (Figure 15). The initial stages show very low enrolment rates for both genders across all education levels, indicating limited access to education. China experienced rapid growth post-1950s, with boys displaying higher enrolment rates at all levels of education. Post-1980s, girls caught up and even surpassed boys in secondary and tertiary education enrolments. This progress in female education is part of a broader trend where China is positioned with relatively high learning outcomes for its GDP per capita compared to other countries (Figure 16). The graph displays the country above the trend line, suggesting better-than-expected educational outcomes given its economic status. However, maintaining high enrolment while addressing quality of education remains critical (Tang, 2023).

Furthermore, for the thesis' purpose, it is necessary to analyse and contextualize women's wellbeing and barriers to entry. As the birth rate declined sharply, influenced by the OCP and economic development, the fertility rate decreased to below replacement level (Figures 17 and 18). Whilst China transitioned from an agrarian to an industrial economy, fertility rates naturally declined due to urbanization, higher living costs, and changing societal norms (Cao, 2022).

⁹ It is important to note that the Gross Enrolment Ratio (GER) – the total enrolment in a specific level of education, regardless of age, expressed as a percentage of the official school-age population for that level – can exceed 100%. This occurs because GER includes all students enrolled, including those who are older or younger than the typical age range for that education level, as illustrated in Figure 14. Factors such as late school entry, early entry, or grade repetition contribute to this phenomenon.

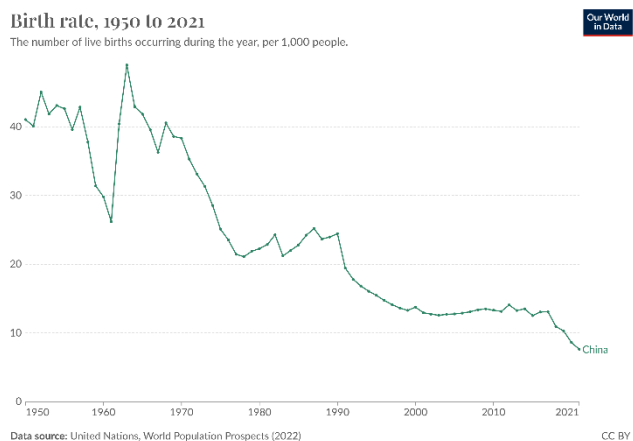


Figure 17. Birth Rate in China (1950-2021).

Source: *Our World in Data* (2024w).

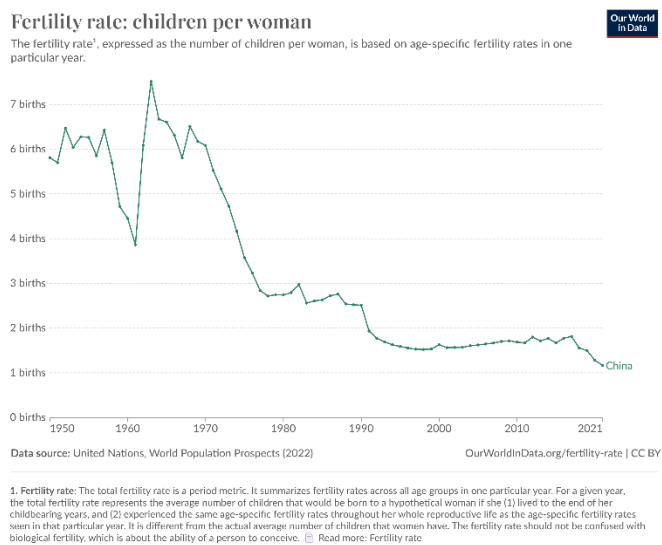


Figure 18. Fertility Rate in China (1950-2021).

Source: *Our World in Data* (2024o).

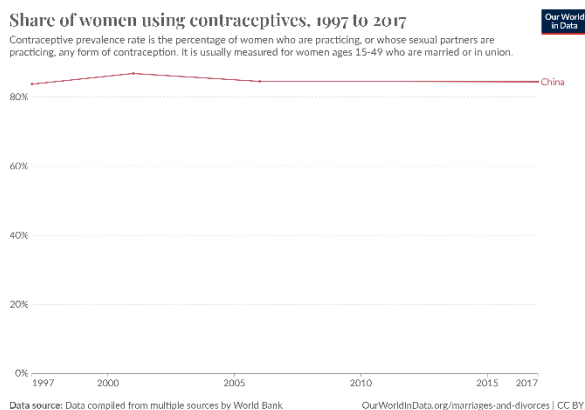


Figure 19. Share of Women Using Contraceptives in China (1997-2017).

Source: *Our World in Data* (2024ae).

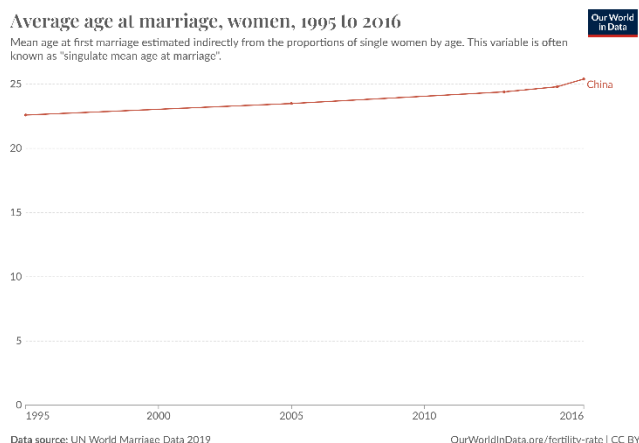


Figure 20. Average Age at Marriage for Women in China (1995-2016).

Source: *Our World in Data* (2024e).

Moreover, the contraceptive prevalence rate has been consistently high, with a slight decline around the early 2000s (Figure 19). The average age at marriage increased from around 23 years in 1995 to 25 years in 2016 (Figure 20), correlating with higher education levels, fewer children, and more career opportunities for women. Urbanization

and changing social norms also contributed to this trend: the SRB has been heavily influenced by cultural preferences and policy changes. Sen's work is particularly relevant to understanding and addressing the skewed gender ratio in China, as he popularized the term "missing women" in an article published in 1990, while highlighting the alarming number of women missing from the population statistics of certain countries, including China and India, due to gender bias and discrimination (Sen, 1990). Not only does gender disparity deny women their rights and opportunities, but it also has wider societal ramifications and seriously jeopardizes social progress and stability. Figure 22 illustrates Bongaarts and Guilmoto's (2015) claim that the number of missing females has increased over the previous few decades, hitting 126 million in 2010, and that number is predicted to peak at 150 million in 2035.

One significant factor observed is that early socialist policies weakened traditional patriarchy in urban areas through industrialization and bureaucratic controls, but rural patriarchy remained strong, reinforcing traditional roles. Yang (2023) suggests that evolving gender beliefs could affect fertility and marriage rates, mirroring trends in post-industrial societies. Addressing gender inequality requires both redistributive policies to ensure equal access to resources and recognition of cultural injustices, since education can perpetuate gender biases if not carefully addressed. Textbooks often contain gender stereotypes, and teachers' awareness of gender discrimination is crucial in mitigating these biases (Cao, 2022).

Sex ratio at birth, 1950 to 2021

The sex ratio at birth is measured as the number of newborn boys for every 100 newborn girls. Higher values indicate a much higher number of newborn boys than girls.

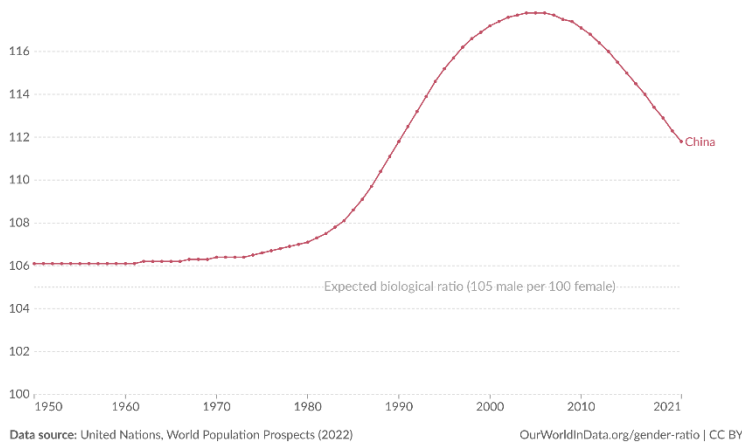


Figure 21. Sex Ratio at Birth in China (1950-2021)

Source: *Our World in Data* (2024ad).

Number of 'missing women' in the world, 1970 to 2050

'Missing women' refers to the number who would be alive in the absence of sex discrimination. Missing women are the sum of women missing at birth (as a result of sex-selective abortion) and excess female mortality through infanticide, neglect or poor treatment.

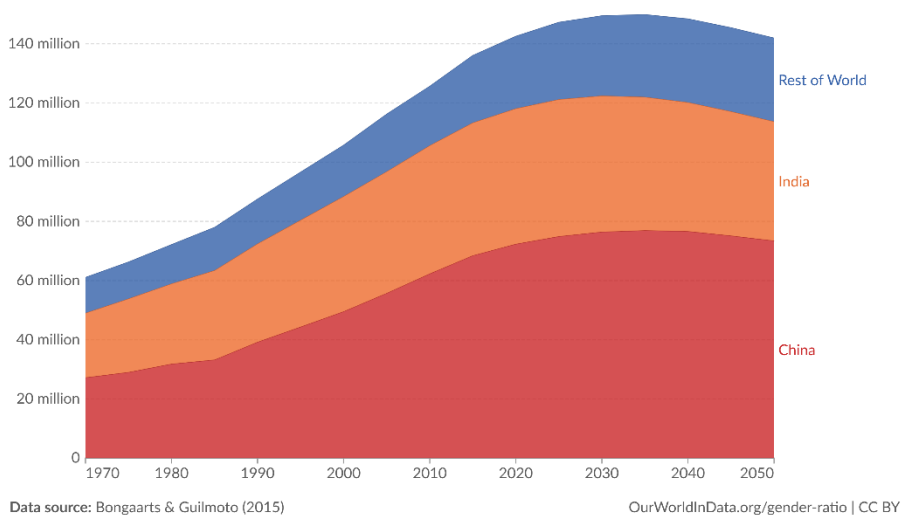


Figure 22. Number of “Missing Women” in the World (1970-2015).

Source: *Our World in Data* (2024v).

Female Labour Participation and Economic Growth in China

It is essential to examine the patterns of female workforce participation in relation to educational attainment to fully understand the dynamics of gender equality within the Chinese labour market. This research showed that gender parity in education by the early 21st century led to more women pursuing higher education and achieving financial independence (Wu, 2022). This has been a crucial factor in rapid economic growth, as evidenced by the increase in GDP per capita from 1970 to 2022 (World Bank, 2022). Furthermore, higher educational attainment among women is associated with better income and empowerment within households, contributing to overall economic development (Ma, 2024). A helpful context for this analysis is provided by the typical U-shaped relation seen in many developing economies, where female labour market participation first decreases with increasing educational attainment before increasing beyond a certain threshold (Verick, 2014). This section will analyse if such a tendency exists in China, with special attention to the consequences for labour productivity, foreign direct investment, and GDP per capita growth (Figure 23 and 24).

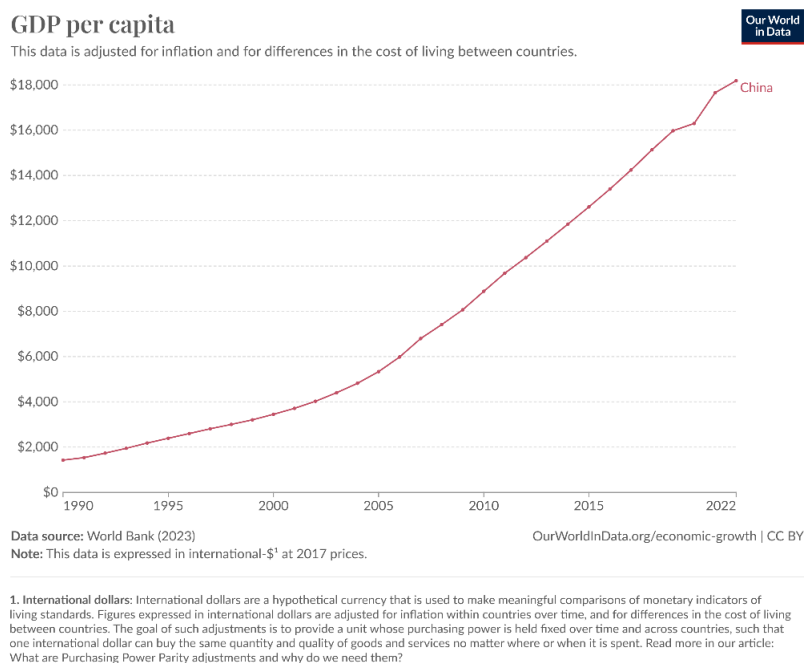


Figure 23. GDP per Capita in China (1990-2022).

Source: *Our World in Data* (2024p).

Annual GDP growth, 1980 to 2029



Annual percent change in gross domestic product¹. This data is adjusted for inflation.



1. **Gross domestic product:** Gross Domestic Product (GDP) is a measure of a country's economic performance. It represents the total monetary value of all goods and services produced within a nation's borders over a specific time period, typically annually or quarterly. GDP includes consumption, government spending, investments, and net exports (exports minus imports). GDP is used to gauge the health of an economy, with increases indicating growth and decreases signaling contraction. Policymakers, economists, and analysts use GDP to make informed decisions and comparisons between countries. It can be measured in nominal terms or adjusted for inflation to reflect real GDP.

Figure 24. Annual GDP Growth in China (1980-2029, with Projected Values)

Source: *Our World in Data (2024d)*.

China's economic growth during the last forty years has been remarkable. Since the economic reforms began, China has become a technical and industrial powerhouse, the second largest economy in the world after the United States (International Monetary Fund, 2024). Chinese GDP grew at an average yearly pace of 9% between 1978 and 2010 (The World Bank, 2024). Major economic changes intended to integrate China into the global economy, along with industrialization and urbanization, drove this rapid expansion. GDP per capita rose dramatically as a result, from \$113.2 in 1970 to about \$12,720.2 by 2022 (World Bank, 2022), reflecting significant improvements in quality of life and financial stability (Figure 23). The growth is particularly steep from the late 1970s onwards, coinciding with the economic reforms initiated, including the introduction of

market mechanisms, opening to foreign trade and investment, and SEZs establishment (Bhardwaj, 1992).

The annual GDP growth rate in China has been highly variable; the initial years post-reform saw extremely high growth rates, often exceeding 10%, reflecting the initial impact of market-oriented restructurings and the low starting base. Throughout the 1990s and early 2000s, growth rates remained robust, typically around 8-10%. China's accession to the WTO in 2001 was a critical milestone that further integrated the country into the global economy, leading to massive growth in trade and investment. Post-2010, growth rates decelerated as China transitioned from an investment-driven model to one reliant on consumption and services, compounded by the 2008-2009 financial crisis. The COVID-19 pandemic also impacted the economy around 2020 (Figure 24). Future projections suggest continued but slower growth, with challenges like debt, an aging population, and sustainable practices becoming more prominent. Environmental sustainability, income inequality (Iwasaki and Ma, 2020), and financial stability are rising concerns. A key component of the country's economic growth has been FDI. China, the largest recipient of FDI, saw inflows peak at \$149.3 billion in 2020 (UNCTAD, 2021), which have enhanced technology transfer, managerial skills, and productivity. Despite significant increases in labour productivity due to industrial upgrading, technology adoption, and education investments, recent years have seen a slowdown, with labour productivity growth dropping from an average annual rate of 9.4% (1978-2018) to about 4% (World Bank, 2020), making it more difficult to maintain rapid economic growth rates.

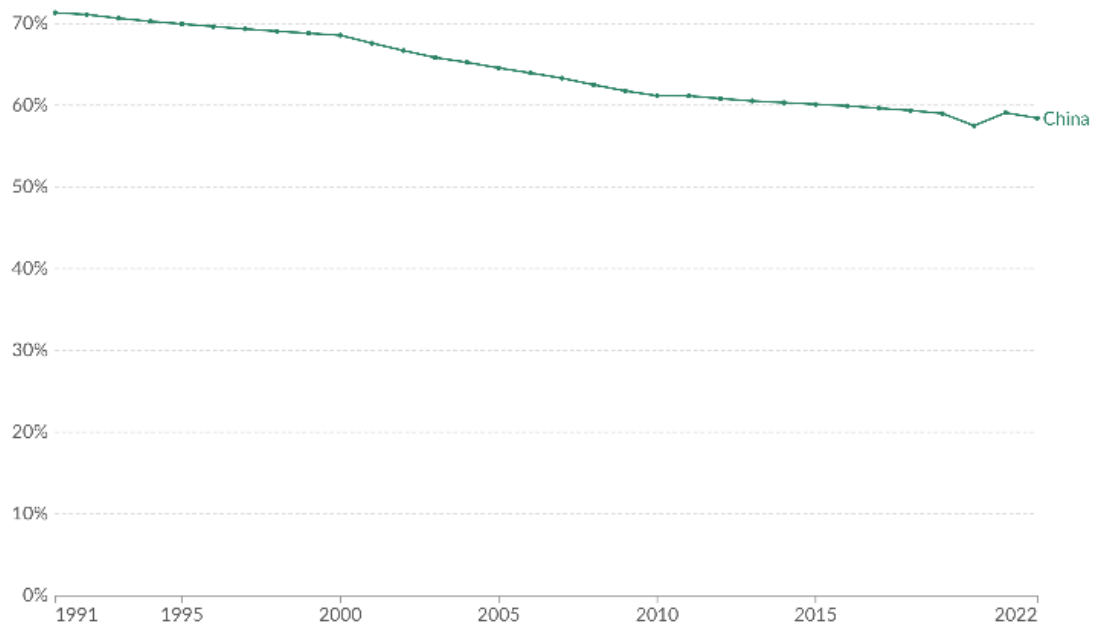
Shifting focus to gender inequalities after the economic status overview, between 1949 and 1977, the Chinese government implemented measures that promoted gender equality and encouraged women's employment during the planned economy era. Both the agricultural and industrial sectors actively integrated women into the workforce, resulting in minimal gender differences in labour force participation (Gustafsson and Li, 2000). However, the 1978 market economy transition disproportionately affected women, who faced higher layoff rates due to state-owned company reorganization (Ngo and Liu, 2016), widening the gender pay gap and reducing FLFP. Consequently, the percentage of women in the workforce has decreased from 73% in 1990 to approximately 61.6% in

recent years (Figure 25). Despite a slight stabilization in women's participation rates due to increased educational attainment, substantial challenges remain, such as balancing childcare responsibilities, the persistent gender pay gap, and a severe shortage of supportive childcare networks (Financial Times, 2023). Therefore, while educational attainment has improved, structural economic transformations have not fully supported continued FLFP, underscoring the need for comprehensive reforms and support systems (Ma, 2018).

Female employment-to-population ratio, 1991 to 2022

The female employment-to-population ratio corresponds to the proportion of a country's female population 15+ that is employed. All figures correspond to 'modeled ILO estimates' (see source for details).

Our World
in Data



Data source: Multiple sources compiled by World Bank (2024) OurWorldInData.org/female-labor-force-participation-key-facts | CC BY

Figure 25. Female Employment-to-Population Ratio in China (1991-2021).

Source: *Our World in Data* (2024m).

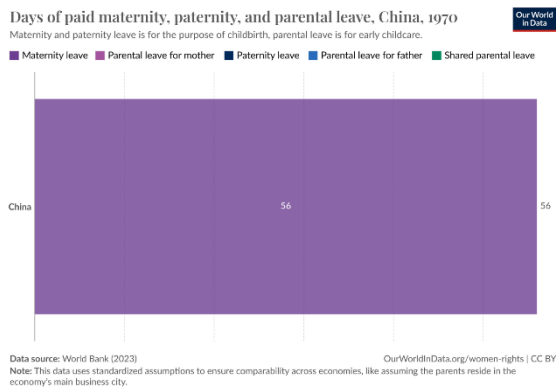


Figure 26

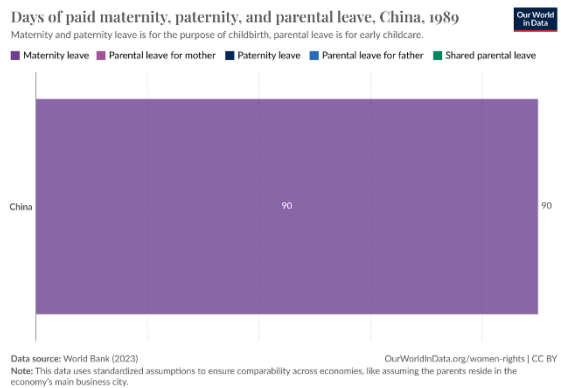


Figure 27

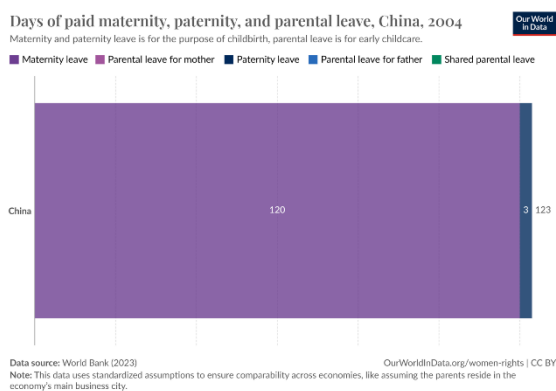


Figure 28

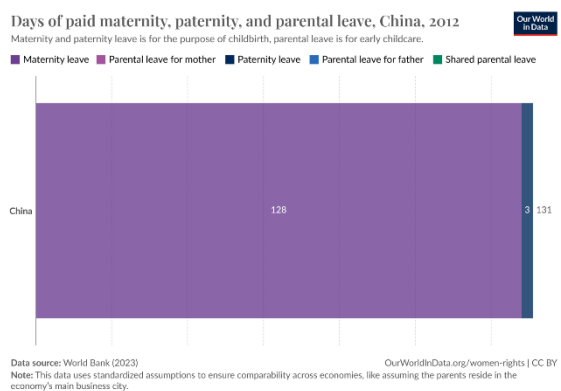


Figure 29

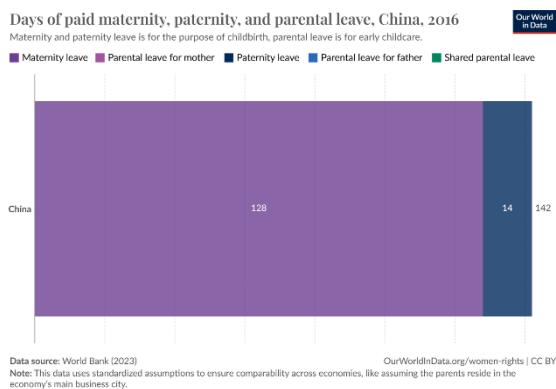


Figure 30

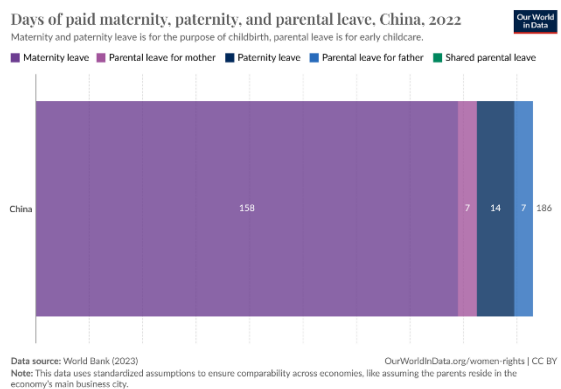


Figure 31

Figures 26 to 31. Longitudinal Analysis of Paid Maternity, Paternity, and Parental Leave Days in China (1970, 1989, 2004, 2012, 2016, 2022).

Source: Our World in Data (2024j).

The evolution of paid maternity, paternity, and parental leave policies in China from 1970 to 2022 signifies a progressive but slow enhancement of parental support systems. In 1970, maternity leave was established at 56 days, a modest provision by contemporary global standards (Figure 26). This period increased steadily to 90 days in 1989 (Figure 27), 120 days in 2004 (Figure 28), 128 days in 2012 and 2016 (Figures 29 and 30), and 158 days by 2022 (Figure 31). Such trajectory highlights the growing recognition of maternal health and early childhood care, aligning with international trends and addressing the negative consequences of the OCP and below-replacement fertility levels. Concurrently, the introduction and gradual increase in paternity leave reflect an evolving understanding of the father's role in childcare and family support. Initially set at 3 days in 2004 (Figure 28), paternity leave increased to 14 days by 2012 (Figure 29) and has since been maintained. This development indicates a shift towards more balanced parental responsibilities, although the leave length for fathers remains considerably shorter than that for mothers. The 2022 introduction of 7 days of parental leave for both parents represents another significant policy shift, acknowledging the extended role of fathers in early childcare (Figure 31).

These policy changes must be contextualized within China's economic growth and social evolution. Economic development has facilitated the capacity to offer extended leave benefits, while increased women's workforce participation and shifting family dynamics have driven the demand for comprehensive parental leave policies. Paternity and parental leaves contribute to shared parenting responsibilities, enhancing family well-being and promoting equality in both workforce and domestic spheres. Since increasing the participation of women in the labour market is essential to maintaining workforce numbers in the future, the government must implement measures to reduce workplace discrimination against women (Iwasaki and Ma, 2020).

Chapter II: Is India's Temperature Rising or Falling?

Initiatives in Indian Education and Gender Equity

This chapter examines the important programs and policies implemented by the Indian government to promote equality and enhance education, focusing on the NIPUN Bharat Scheme, Sarva Shiksha Abhiyan, Beti Bachao Beti Padhao Scheme, and National Education Policy 2020. These initiatives address historical and socioeconomic inequalities that have hampered women's access to education and empowerment. The NEP 1986 brought a significant shift in educational policy, recognizing education as an instrument for women's empowerment. The Programme of Action in 1992¹⁰ delineated further standards for empowerment, including self-confidence, communal unity, equitable societal involvement, and financial autonomy. The establishment of the National Commission for Women in 1992¹¹ demonstrated the government's commitment to addressing female constitutional and legal rights and suggesting legislative changes to improve their opportunities.

¹⁰ The Programme of Action was an elaborate implementation plan introduced by the Government of India to operationalize the principles set forth in the National Education Policy 1986. It aimed to reform the educational system comprehensively and address the diverse needs of different social groups, with a significant emphasis on women's education and empowerment. The document contained 23 chapters, each focusing on specific aspects of education, including education for women's equality, education of scheduled castes, scheduled tribes and other backward sections, minorities' education, education of the handicapped, adult and continuing education, early childhood care and education, elementary education, secondary education, Navodaya Vidyalayas, vocational education, higher education, open education, delinking degrees from jobs and manpower planning, rural universities and institutes, technical and management education, research and development, the cultural perspective, development of languages, media and educational technology, sports, physical education and youth, evaluation process and examination reforms, teachers and their training, and the management of education (Ministry of Human Resource Development, 1992). Each chapter aimed to improve both the content and process of education, making the system more effective and responsive to the needs of different groups. The POA was instrumental in laying down a strategic roadmap for educational development in India and advocated for the integration of gender-sensitive curricula, the training of educators to be more responsive to the needs of female students, and the establishment of monitoring mechanisms to ensure the effective implementation of these measures.

¹¹ The National Commission for Women was established in January 1992 under the National Commission for Women Act, 1990 (National Commission for Women, 2024). It was created to review constitutional and legal safeguards for women, recommend legislative measures, facilitate grievance redressal, and advise the government on policy matters affecting women. The idea for the NCW was initially proposed by the Committee on the Status of Women in India nearly two decades earlier. Various committees and the National Perspective Plan for Women (1988-2000) also recommended its formation. After consultations with NGOs, social workers, and experts, the bill was introduced in May 1990, amended to include civil court powers, and passed in August 1990.

The 1990s saw an increased focus on compulsory education to empower women through initiatives such as the National Policy for the Empowerment of Women¹², formulated by the Indian Department of Women and Child Development. Notwithstanding notable advancements, obstacles like socioeconomic divides, cultural norms, and geographical differences still prevent many objectives from being achieved. For instance, the ASER (Ministry of Finance, 2020) noted that while female school enrolment rates have increased, significant disparities persist in attendance and retention, especially in rural areas. As the concept of loss aversion, introduced by Kahneman and Tversky (1979), suggests, people prefer avoiding losses to acquiring equivalent gains: in the context of girls' education in India, parents might fear losing immediate benefits, such as their daughters' labour at home or the dowry advantages of early marriage.

National Education Policy 2020

NEP 2020 was developed following the overarching goal of modernising Indian education to match 21st-century requirements. Approved by the Union Cabinet on July 29, 2020, is the first major education policy revision in 34 years, since the 1986 version (Kumar, 2021). The Right to Education Act, under Article 21A of the Indian Constitution (2010) stipulates that “[T]he State shall provide free and compulsory education to all children of the age of six to fourteen years in such manner as the State may, by law, determine [...]” without regard to caste or gender. Following these regulations, the policy is built on accessibility, equity, affordability, quality, and accountability (Ministry of Human Resource Development, 2020). NEP 2020 lays out a goal to guarantee all students fair access to high-quality education and to cultivate an inclusive learning environment that is cognizant of a range of backgrounds and academic skill levels; by embracing technology, it places an emphasis on encouraging critical thinking, problem-solving

¹² The National Policy for the Empowerment of Women, introduced by the Indian government in 2001, aims to promote women's advancement and development in various sectors. This policy focuses on creating supportive economic and social conditions that allow women to realize their full potential (Ministry of Women and Child Development, 2010). It advocates for equal enjoyment of human rights and fundamental freedoms for women, ensuring their participation in decision-making processes in political, economic, and social spheres. The policy also emphasizes equal access to health care, education, and employment opportunities, along with fair remuneration and social security. It seeks to strengthen legal frameworks to eliminate discrimination and violence against women, promote gender-sensitive societal attitudes, and foster partnerships with civil society and women's organizations.

abilities, and experience learning in place of memorization (ibid.). It aims to facilitate a smooth transition from early childhood to higher education through comprehensive, learner-centred experiences (Aithal and Aithal, 2020). Furthermore, emphasizing Early Childhood Care and Education, the policy aims for universal and quality access for children aged three to six by 2030 through a network of Anganwadis¹³, preschools, and kindergartens (Ministry of Human Resource Development, 2020). Moreover, the emphasis on FLN is a crucial aspect of NEP 2020 (Kumar, Pratap and Aggarwal, 2023). The policy sets an ambitious goal: by 2025, all elementary schools will have universal FLN, with an emphasis on fundamental reading, writing, and math skills at the primary school level, by involving parents, teachers, and community members. A number of educational changes are implemented in order to improve student learning results and lower the dropout rate: the 10+2 structure is replaced by a 5+3+3+4 design,

“[...] consisting of the Foundational Stage (in two parts, that is, 3 years of panchwadi/pre-school + 2 years in primary school in Grades 1-2; both together covering ages 3-8), Preparatory Stage (Grades 3-5, covering ages 8-11), Middle Stage (Grades 6-8, covering ages 11-14), and Secondary Stage (Grades 9-12 in two phases, i.e., 9 and 10 in the first and 11 and 12 in the second, covering ages 14-18)” (Ministry of Human Resource Development, 2020, p.11).

Additionally, the strategy also emphasized research from the undergraduate level holistic student development through outcome-based learning and cross-disciplinary education. The National Academic Bank of Credit will allow flexible degree attainment by enabling credit transfers between institutions (Kumar, 2021). NEP 2020 places a strong emphasis on the private sector's expanded involvement in education. Based on their accreditation status, involvement in research, and calibre of instruction, private HEIs will be awarded varying degrees of autonomy. These educational establishments must uphold open and honest procedures; penalties, including possible closure, may follow noncompliance with the rules and criteria (Kumar, Pratap and Aggarwal, 2023). The digital divide, opposition to change, and regional imbalances are some of the obstacles that NEP 2020 must overcome to achieve its objectives. Reducing teacher shortages (Tang, 2023) and promoting multilingual education, especially in areas where local languages predominate, are critical for success (Kumar, Pratap, and Aggarwal, 2023).

¹³ Anganwadi workers provide basic health care and pre-school education in rural areas (India Development Review, 2019).

Overall, NEP 2020 is on the right path towards achieving the SDGs, incentivizing lifelong and sustainable learning with flexible entry-exit points for degree completion.

Beti Bachao Beti Padhao Scheme

Gender inequality in India, rooted in socioeconomic, cultural, and historical factors, manifests in biased sex selection, lower female literacy rates, restricted access to healthcare and employment for women, an unbalanced CSR and widespread prejudice against females. To address these issues, Prime Minister Narendra Modi introduced the BBBP¹⁴ programme at Panipat, Haryana, on January 22, 2015 (Sharma and Pattanayak, 2022). The scheme aims to prevent gender-based sex-selective elimination, ensure the survival and protection of female children, and to guarantee that girls have equal access to high-quality education and opportunities for both personal and professional growth (Ministry of Women and Child Development, 2015). India has enacted several laws to empower women and girls, including the previously mentioned RTE, the Prohibition of Child Marriage Act¹⁵, and the Pre-Conception and Pre-Natal Diagnostic Techniques

¹⁴ The phrase “Beti Bachao, Beti Padhao,” commonly translated in English as “Save the Daughter, Educate the Daughter,” encapsulates the dual-objective social campaign launched by the Government of India in January 2015. The campaign’s title addresses two significant societal issues. The first part, “Beti Bachao” (बेटी बचाओ), translates directly from Hindi as “Save the Daughter,” highlighting the urgent need to protect female children from female foeticide and infanticide, practices driven by deeply entrenched gender biases and a preference for male offspring, societal and systemic threats to their survival and well-being. The second part, “Beti Padhao” (बेटी पढ़ाओ), translates to “Educate the Daughter.” This element emphasizes the importance of providing equal educational opportunities to girls, which is essential for promoting gender equality and empowering women. By advocating for the education of female children, the campaign aims to address the disparities in educational attainment and ensure that girls have the same opportunities for personal and professional development as boys. The campaign was initiated in response to the alarming decline in the child sex ratio, which pointed to a significant gender imbalance exacerbated by sex-selective abortions and societal neglect of girls. By articulating these two interconnected goals – protecting and educating – the slogan “Beti Bachao, Beti Padhao” effectively raises awareness and mobilizes public support: the simplicity and directness of the phrase contribute to its impact, making it a potent tool for social change across diverse segments of the Indian population.

¹⁵ The Prohibition of Child Marriage Act, 2006, enacted by the Government of India, serves as a crucial legislative instrument aimed at eradicating the practice of child marriage and safeguarding the rights of minors (Government of India and Ministry of Women and Child Development, 2006). The Act is structured into 21 sections and has a comprehensive jurisdiction across India. The Act’s provides precise definitions essential for the implementation and enforcement of its provisions. A “child” is defined as a male who has not attained the age of 21 years or a female who has not reached the age of 18. The term “contracting party” pertains to any individual whose marriage is either solemnized or is on the verge of being solemnized under this Act. “Child marriage” is defined as a marriage in which either party meets the criteria of a child as per the Act’s definitions. Additionally, the term “minor” refers to an individual who has not yet achieved the status of majority as delineated by the Indian Majority Act. Any marriage involving a child is voidable at the discretion of the contracting party who was underage at the time of the marriage ceremony. This individual is entitled to seek a decree of nullity by submitting a petition to the district court within two years

Act¹⁶. These laws form the cornerstone for social and legal transformation, which are essential to advance the goals of the BBBP scheme. Notwithstanding the progress since its launch, the programme still has a way to go before reaching its goals. Coordination among the Ministry of Women and Child Development, the Ministry of Health and Family Welfare, and the Ministry of Human Resource Development poses difficulties due to administrative gaps and lack of coordination. Moreover, strict and effective enforcement by police and government agencies, often lacking in many regions, is crucial for the scheme's success, necessitating full civic society support, as the influence of the project has been diminished due to the community members' negligible involvement (Meenakshi and Bakshi, 2022). Cultural perceptions of girls need to change through active community participation, as ingrained conventions hamper progress despite awareness campaigns and orientation programmes under BBBP. The persistence of female foeticide and dowry customs undermines the scheme's objectives, thus enforcing stronger legislation is increasingly necessary. Parents' reluctance to send daughters to school due to crime and safety concerns also presents a barrier, necessitating the creation of a safe and informed environment for girls' education. However, the benefits of the Scheme are not well known to health workers in certain rural regions, including ASHA¹⁷, Anganwadi, and ANM staff¹⁸. Hence, further programmes for sensitization and effective training are required (Meenakshi and Bakshi, 2022). Local government structures, such as Panchayats¹⁹ and other urban local bodies are vital for BBBP's implementation, as they oversee and provide support, guaranteeing accountability and community involvement at the local level, and putting laws and regulations into effect and upholding them to forward the programme's goals. Notable accomplishments since its inception include female

of attaining the age of majority. Furthermore, upon issuing such a decree, the district court is obliged to mandate the restitution of money, gold, ornaments, gifts, and other valuables exchanged during the marriage to the respective parties and their parents or guardians

¹⁶ The Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994, was enacted to stop female foeticides and improve the declining sex ratio in India. The Act bans prenatal sex determination and regulates genetic counselling centres, laboratories, and clinics involved in pre-natal diagnostic techniques, including IVF and preimplantation genetic diagnosis (Government of India, 1994).

¹⁷ ASHA (Accredited Social Health Activist) focus on promoting health education and facilitating access to healthcare services (India Development Review, 2019).

¹⁸ ANM (Auxiliary Nurse Midwife) staff are community health workers in India. They offer maternal and child health services (India Development Review, 2019).

¹⁹ Panchayats are local self-government institutions in rural India, functioning at the village (Gram Panchayats), intermediate (Panchayat Samitis), and district (Zilla Parishads) levels. They are responsible for local administration and rural development, promoting grassroots democracy and community participation.

higher enrolment in schools, improved SRB, and shifts in societal perceptions of girls, which are the result of creative efforts like Udaan²⁰, My Aim My Target Campaign²¹, Digital Guddi-Gudda Board²², Lakshya Se Rubaru²³ and others (India Development Review, 2019). However, ongoing challenges include monitoring effectiveness, cultural resistance, and implementation gaps (Meenakshi and Bakshi, 2022). To overcome these obstacles, a behavioural change seems required.

Sarva Shiksha Abhiyan

SSA was introduced in 2000-2001 with the goal of achieving universal access to basic education, focusing on retention, closing social and gender gaps, and raising student achievement levels. The SSA involves a wide range of actions implemented in collaboration with state governments (Yadav, Sharma, and Birua, 2018). These interventions comprise establishing new schools, building additional classrooms, offering alternative education, ensuring separate restrooms and drinking water facilities, providing free textbooks, uniforms, and other materials, and offering teacher training and academic resource support (Draboo, 2020). A primary objective is to grant education to all, whether through conventional schools, Education Guarantee Centres, Alternative Schools, or Back-to-School camps, ensuring all children aged six to fourteen complete eight years of elementary schooling (Kainth, 2006).

Launched in 2004 and merged with SSA in August 2007, the KGBV²⁴ scheme aims to establish residential upper primary schools for girls from marginalized and

²⁰ Udaan - Sapne Di Duniya De Rubaru is an initiative aimed at providing young girls with the opportunity to shadow professionals in their fields of interest, thereby facilitating career exploration and exposure (India Brand Equity Foundation, 2024).

²¹ The My Aim My Target Campaign is a recognition programme that awards and celebrates the top academic achievements of girls in higher secondary schools, thereby encouraging academic excellence (India Brand Equity Foundation, 2024).

²² The Digital Guddi-Gudda Board serves as a digital platform to highlight gender disparities in birth rates and disseminate information on various schemes and programmes designed to protect and promote the welfare of the girl child (India Brand Equity Foundation, 2024).

²³ Lakshya Se Rubaru is an internship programme specifically tailored for female college students, with the goal of enabling them to make informed career decisions through practical experience and guidance (India Brand Equity Foundation, 2024).

²⁴ Kasturba Gandhi Balika Vidyalaya is a scheme launched by the Government of India in 2004 to provide educational facilities for girls from marginalized communities, including Scheduled Castes, Scheduled Tribes, Other Backward Classes, and minority groups. KGBV schools offer residential schooling to ensure

disadvantaged communities (Draboo, 2020). These schools are set up in Educationally Backward Blocks (EBBs) where female literacy is below the national average and the gender gap in literacy exceeds the national average, “with a female literacy rate below 46 percent and a gender gap higher than 23 percent” (The World Bank, 2017, p.6). Seventy-five percent of the seats in the KGBV plan are reserved for girls from SC²⁵, ST²⁶, and other minority communities, with the remaining 25% for girls from families below the poverty level (Draboo, 2020). In order to secure girls’ economic independence and skill development, KGBVs offer free education, residential amenities, and vocational training.

Furthermore, a unique gender component plan under the SSA, the NPEGEL, aims to raise the educational attainment of poor girls from marginalised communities aged six to fourteen (Draboo, 2020). The program establishes model cluster schools in all districts and blocks with active community involvement. These model schools support ECCE, distribute gender-sensitive teaching and learning resources, and provide need-based incentives such as escorts for girls, stationery, and uniforms (Ministry of Human Resource Development, 2003). NPEGEL strengthen the supply side of education by enhancing educational facilities and creates a supportive learning environment, keeping girls in school by providing resources like sports equipment, computer-aided instruction, and career training. All things considered, SSA has significantly improved girls’ education in India since its founding. Girls now make up a larger portion of the student body in elementary schools, and efforts to lower the dropout rate have had a favourable impact on the number of female students who complete their elementary education.

NIPUN Bharat Scheme

The Government of India announced the National Initiative for Proficiency in Reading with Understanding and Numeracy with the goal of guaranteeing that all children

a supportive learning environment, addressing barriers such as distance from schools, safety concerns, and socio-economic challenges that often hinder girls’ education.

²⁵ According to Article 366 (24) of the Constitution of India (1949), the term “Scheduled Castes” is defined as: “Such castes, races, or tribes or parts of or groups within such castes, races, or tribes as are deemed under Article 341 to be Scheduled Castes for the purposes of this Constitution.”

²⁶ Article 366 (25) of the Constitution of India (1949) defines “Scheduled Tribes” as: “Such tribes or tribal communities or parts of or groups within such tribes or tribal communities as are deemed under Article 342 to be Scheduled Tribes for the purposes of this Constitution.”

have a solid foundation in reading and numeracy by 2026–2027. Announced on July 5, 2021, NIPUN Bharat²⁷ is part of the centrally funded Samagra Shiksha programme²⁸ (Sharma and Pattanayak, 2022). Its primary goal is to ensure all Indian children complete Grade III or Grade V with essential reading, writing, and numeracy skills (ibid.). These skills are crucial for continued education and have widespread benefits, especially for SC and ST females and urban children (Purpura, Litkowski and Knopik, 2019). To accomplish its goals, NIPUN Bharat concentrates on activity-based learning, the use of interactive and engaging teaching strategies to improve comprehension and retention in early learners, and the creation and implementation of curriculum that is age-appropriate – that is, appropriate for the cognitive and developmental stages of young children (Ministry of Education, 2021). It also provides mentorship and support networks for teachers to improve their methods and offers opportunities for ongoing professional development (ibid.).

Caste remains a powerful social force affecting a person’s whole life, including educational opportunities (Borooah and Iyer, 2005). Discrimination based on caste and gender negatively impacts the engagement and academic achievement of students from underprivileged backgrounds, correlating with higher dropout rates (Paik, 2014). Thus, it is imperative that oversight bodies take proactive measures to stop this kind of prejudice to keep students and advance fair education.

NEP 2020, BBBP, SSA, and NIPUN Bharat have combined influence on changing India’s educational landscape and contribute to economic growth by creating a skilled and capable workforce. Improved educational outcomes, indicated by higher enrolment rates, correlate with better job prospects and economic stability, thus driving economic growth (ASER, 2023). This chapter provided the policy and initiative’s foundation needed to set the stage for the following analysis of educational trends and results impacting gender equality and economic development.

²⁷ “Bharat” is the name for India in Hindi.

²⁸ Samagra Shiksha is an overarching programme for the school education sector in India, extending from pre-school to class 12. It was introduced in the Union Budget of 2018-19 with the aim of treating school education holistically, without segmentation from pre-nursery to class 12. The programme aims to improve school effectiveness by ensuring equal opportunities for schooling and equitable learning outcomes. It integrates three previously separate schemes: Sarva Shiksha Abhiyan, Rashtriya Madhyamik Shiksha Abhiyan, and Teacher Education (Department of School Education and Literacy, 2020).

Examining Trends and Gender Equality in India: Impact and Challenges

This section explores the trends and patterns in education and gender equality in India, examining current progress and challenges. It employs a combination of historical context, recent statistical data and figures retrieved by Our World in Data, and analysis of the previously mentioned government policies and societal factors impacting the results observed.

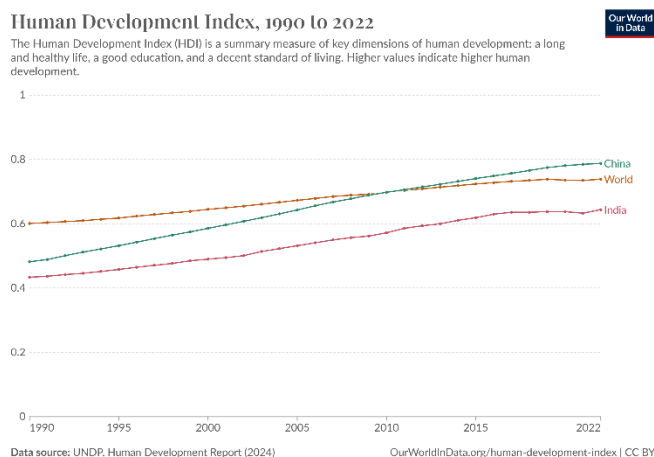


Figure 32. Comparative Human Development Index Trends for India, China, and the World Average (1990-2022).

Source: *Our World in Data* (2024t).

India, classified under medium human development, has shown consistent improvement (Figure 32). By the early 2000s, the HDI exhibited significant progress, thanks to initiatives such as the National Rural Health Mission²⁹ and SSA (Ministry of Health and Family Welfare, 2005; Ministry of Human Resource Development, 2003). Despite starting with a lower HDI than the world average, it has gradually closed the gap: efforts to address gender inequalities began in the 1990s, focusing on female health, education, and economic empowerment through initiatives like BBBP (Ministry of Women and Child Development, 2015). These programs have led to increased female school enrolment and improved maternal healthcare services. In the early 1990s, a high

²⁹ The National Rural Health Mission (NRHM) was launched by the Government of India in 2005 to improve healthcare delivery across rural areas. Key goals of the NRHM include the reduction of the Infant Mortality Rate (IMR) and Maternal Mortality Ratio (MMR), universal access to public health services (covering women's health, child health, water, sanitation and hygiene, immunization, and nutrition), prevention and control of both communicable and non-communicable diseases (including locally endemic diseases), access to integrated and comprehensive primary healthcare, population stabilization, gender and demographic balance, revitalization of local health traditions, and mainstreaming of AYUSH (Ayurveda, Yoga, Unani, Siddha, and Homeopathy) (Ministry of Health and Family Welfare, 2005). The mission also promotes healthy lifestyles to enhance the overall well-being of the rural population (ibid.).

GII score revealed significant disadvantages for women in reproductive health, empowerment, and economic status. Consequently, the 1990s and 2000s saw numerous initiatives to address these inequalities.

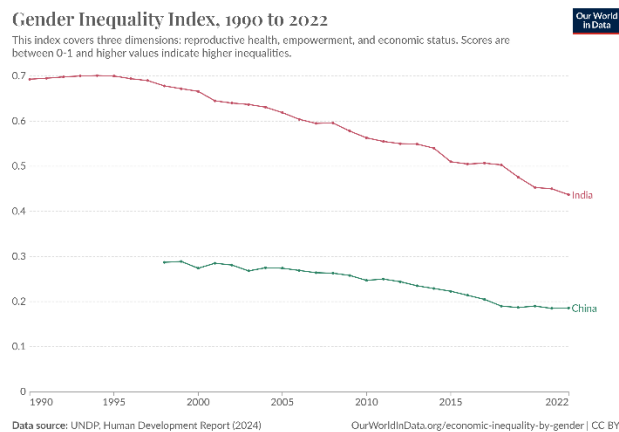


Figure 33. Gender Inequality Index Comparison Between India and China (1990-2022).

Source: *Our World in Data (2024s)*.

Therefore, government policies focused on improving maternal health, such as the Janani Suraksha Yojana³⁰ (Ministry of Health and Family Welfare, 2005), were implemented and played a crucial role in reducing maternal mortality rates. Attempts to increase women's participation in politics and the workforce, supported by legal reforms and economic policies, began to reshape Indian landscape. Indeed, by the mid-2000s, the GII showed signs of improvement: women's literacy rates were on the rise, and more women were entering higher education and professional fields (Drèze and Sen, 2013). However, the score still lags behind countries like China, despite the incremental positive improvements (Figure 33).

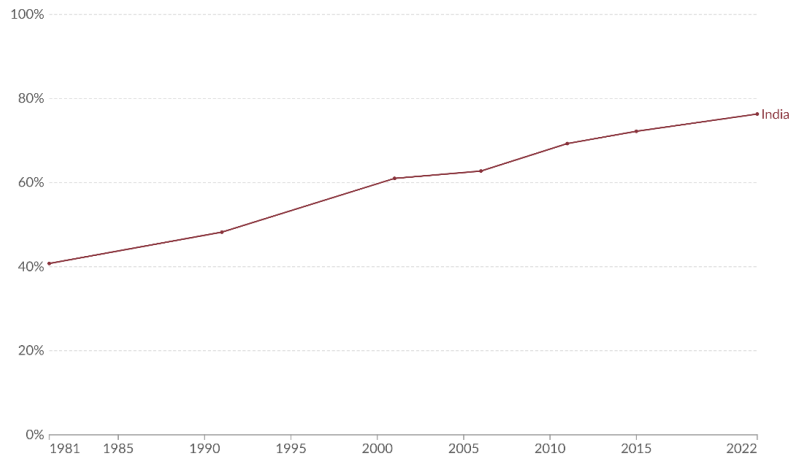
In the context of education trends and patterns, the colonial era reoriented the educational focus towards creating a workforce suited for administrative tasks. This switch significantly compromised the quality and inclusivity of education. Nonetheless, in the post-independence era, education was reconceptualized as a key instrument for fostering a better society (Kumar, 2021). The literacy rate improved significantly (Figure 34), reflecting the establishment of numerous institutions and policies aimed at

³⁰ Janani Suraksha Yojana is a motherhood protection scheme under the National Health Mission in India, launched in 2005. Its primary objective is to reduce maternal and neonatal mortality by promoting institutional delivery among poor pregnant women through financial incentives. It also provides incentives to accredited social health activists (ASHAs) for supporting pregnant women during pregnancy and childbirth.

universalizing education, such as the RTE and various national missions. In 1981, the literacy rate for adult women was slightly above 20% (Figure 37), and by 2018, it had increased to over 60%. The continuous upward trend indicates and confirms the ongoing improvements in educational access and quality for adult women. Starting at around 50% in 1981, the literacy rate in men has climbed to approximately 80% by 2018, showing an important gender gap (Figure 35). However, there has been a steady increase of the female youth literacy rate over the years, reaching nearly 80% by 2022 (Figure 36), confirming the progress in female education and literacy efforts generation after generation. Indeed, from the 1980s onwards, the average years of schooling increased exponentially and the analysis of average years of schooling by gender reveals the evolution of the gender discrimination over time (Figure 39). By 2022, the gender gap had significantly narrowed, indicating positive impacts of sustained policy efforts. However, these data do not explain reality in a thorough manner: women, particularly in the patriarchal northern regions, often face discrimination, limiting access to food, sanitation, and education (Hooda, 2021). Moreover, girls' dropout rates are influenced by early marriages, domestic responsibilities, and lack of nearby schools. According to Hooda (2021), while the gender gap in education is narrowing, significant inequalities remain, especially in rural areas.

Literacy rate, 1981 to 2022

The share of adults aged 15 and older who can both read and write.



Data source: World Bank (2023); Various sources (2018)

OurWorldInData.org/literacy | CC BY

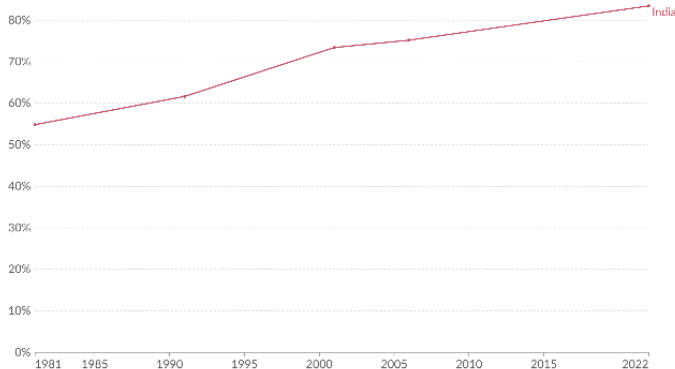
Note: Specific definitions and measurement methodologies vary across countries and time.

Figure 34. Literacy Rate in India (1981-2022).

Source: *Our World in Data* (2024a).

Literacy rate in adult men, 1981 to 2022

Percentage of men aged 15 years and older who can both read and write.



Data source: Multiple sources compiled by World Bank (2024)

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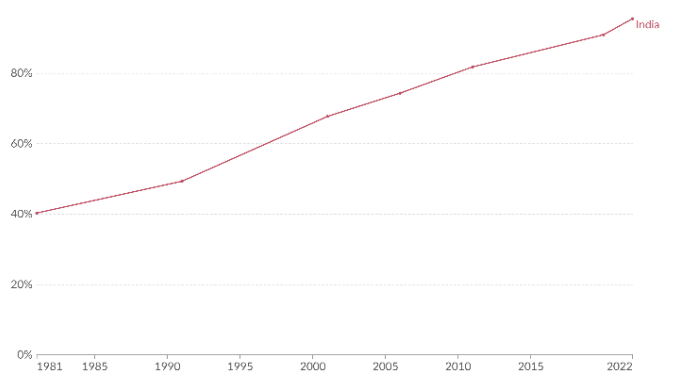
Note: UNESCO defines someone as literate who can read, understand, and write a short, simple statement about their everyday life. But definitions and criteria can vary by country.

Figure 35. Literacy Rate Among Adult Men in India (1981-2022).

Source: *Our World in Data* (2024c).

Female youth literacy rate, 1981 to 2022

Share of women aged between 15 and 24 years old who can both read and write.



Data source: UNESCO Institute for Statistics via World Bank (2023)

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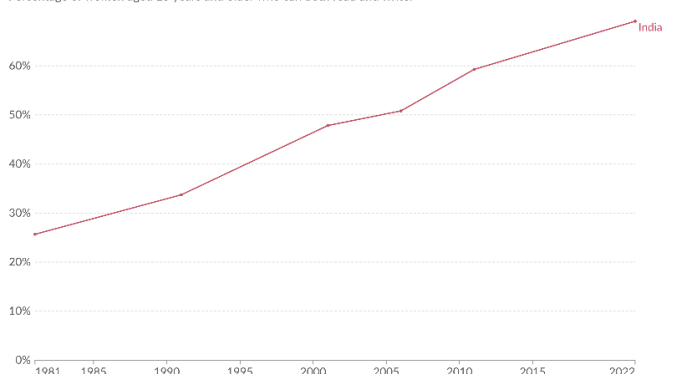
Note: The UNESCO defines a literate adult as someone who can, with understanding, read and write a short, simple statement on their everyday life.

Figure 36: Female Youth Literacy Rate in India (1981-2022).

Source: *Our World in Data* (2024n).

Literacy rate in adult women, 1981 to 2022

Percentage of women aged 15 years and older who can both read and write.



Data source: Multiple sources compiled by World Bank (2024)

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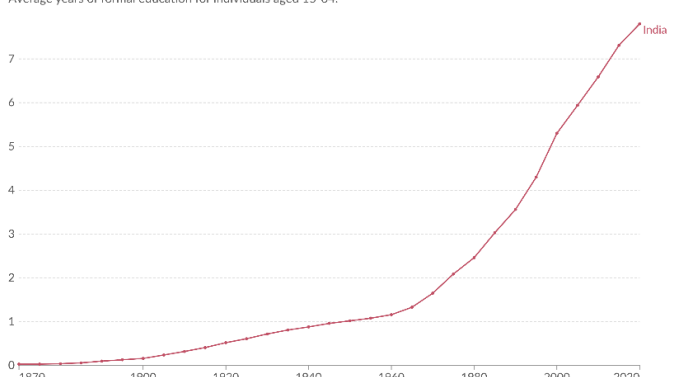
Note: UNESCO defines someone as literate who can read, understand, and write a short, simple statement about their everyday life. But definitions and criteria can vary by country.

Figure 37. Literacy Rate Among Adult Women in India (1981-2022)

Source: *Our World in Data* (2024b)

Average years of schooling

Average years of formal education for individuals aged 15-64.



Data source: Barro and Lee (2015); Lee and Lee (2016)

OurWorldInData.org/global-education | CC BY

Note: Formal education is primary¹/ISCED 1 or higher. This does not include years spent repeating grades. Data for the years before 2015 are estimates, while data from 2015 onwards are projections.

1. Primary education: Primary education (International Standard Classification of Education Level 1) aims to impart fundamental literacy and numeracy skills while providing a solid foundation in key knowledge areas and personal and social development, serving as preparation for lower-secondary education with a focus on basic-level learning and minimal specialization.

Figure 38. Average Years of Schooling in India (1870-2020).

Source: *Our World in Data* (2024h).

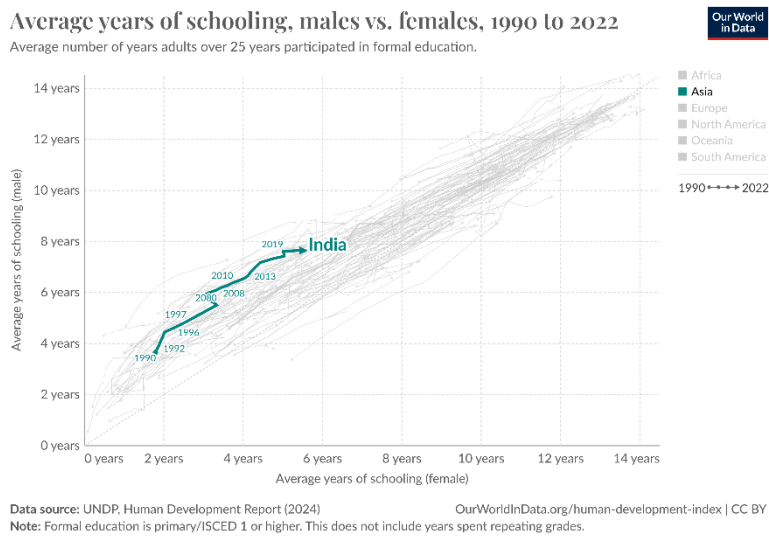


Figure 39. Average Years of Schooling by Gender in India (1990-2022).

Source: *Our World in Data* (2024i).

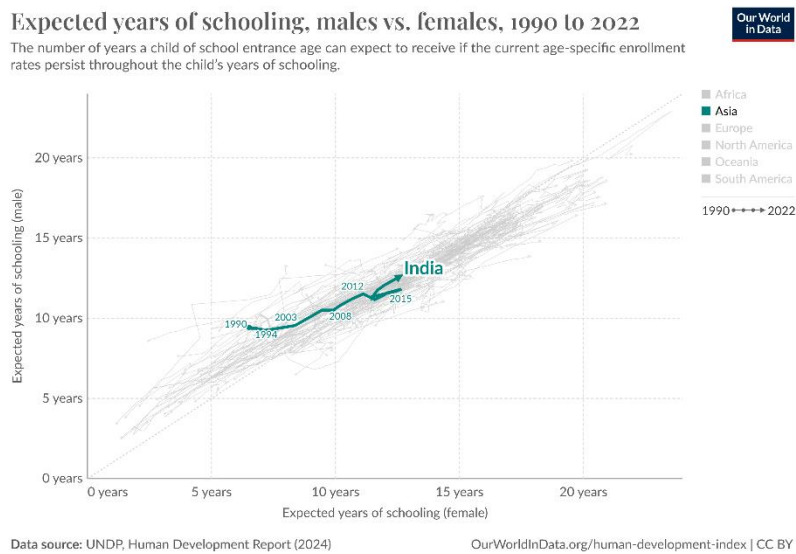


Figure 40. Expected Years of Schooling by Gender in India (1990-2022).

Source: *Our World in Data* (2024l).

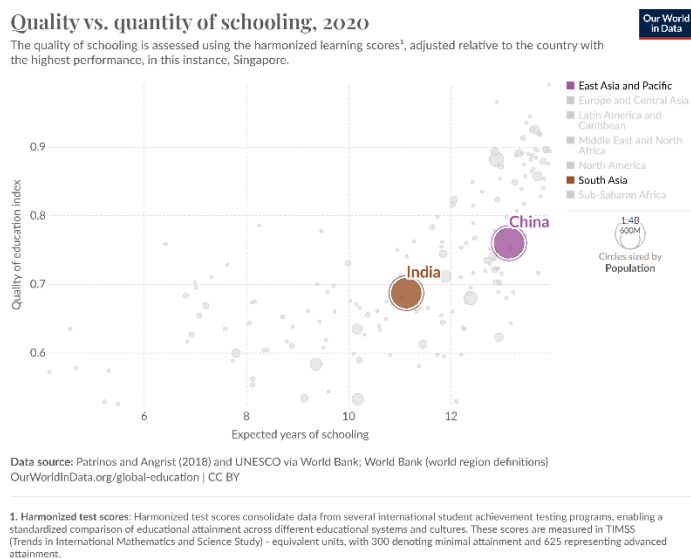


Figure 41. Comparative Analysis of Quality vs. Quantity of Schooling in India and China (2020).

Source: *Our World in Data* (2024ab).

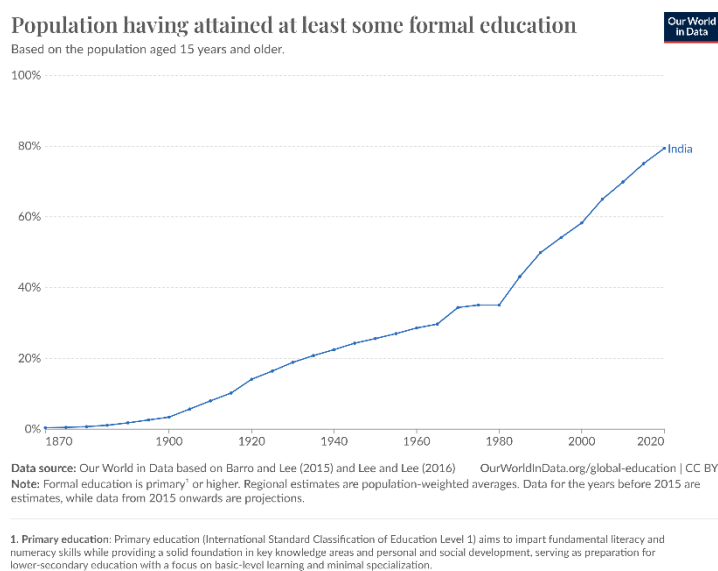


Figure 42. Population with at Least Some Formal Education in India (1870-2020).

Source: *Our World in Data* (2024x).

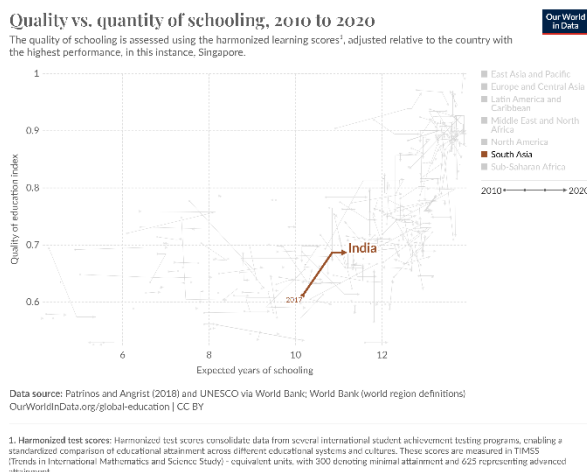


Figure 43. Quality vs. Quantity of Schooling in India (2010-2020).

Source: *Our World in Data* (2024ab).

All in all, the data indicates improvements in education quantity, with higher enrollment rates and longer retention in the education system, thanks to efforts to reduce dropouts rates, such as midday meal schemes (Jayaraman and Simroth, 2015) and conditional cash transfers (Sekher and Ram, 2015). According to the World Population Review (2021), India's quality of education index is approximately 0.55. This relatively lower score compared to China underscores significant challenges in quality and accessibility of education, including inadequacies in teacher training, infrastructure, and outdated curricula (Jha and Parvati, 2014). Interestingly, according to the Annual Status of Education Report 2022 (ASER, 2023), the proportion of children aged six to fourteen enrolled in government schools in rural areas markedly increased from 65.6% in 2018 to 72.9% in 2022. Historically, as Figure 42 shows, formal education was nearly non-existent in 1870 but rose to 15% by 1950 and 40% by 1980. From 2000 to 2020, the growth rate of educational attainment surged, with projections indicating over 80% of the population having formal education by 2020. Notably, primary enrolment rates have remained consistently high, reaching nearly 100% by 2022 (Figure 44). Lower secondary completion rates increased from below 40% in 1971 to around 70% by 2022, reflecting improved access to secondary education. Tertiary enrolment grew from a minimal base

in 1971 to approximately 30% by 2022, as shown (Figure 44). India now boasts the world's largest school education system, with around 15 lakhs³¹³² schools, 96 lakhs³³ teachers, and 26 crores³⁴³⁵ students (Ministry of Education, 2022). Moreover, the Gross Access Ratio, reflecting school availability based on prescribed norms, stands at 100.13% per elementary and 79.56% for secondary levels, with primary dropout rate around 1.45% and secondary around 12.61% (ibid.). The enrolments of both boys and girls in primary education began to rise significantly around the year 1900 (Figure 45). By 1950, the rate for boys was almost two times higher. Afterwards, a marked increase in enrolment is observed from 1950 to 2000, with both genders reaching around 80% enrolment by the end of this period. By 2021, the enrolment rates for both boys and girls in primary education had nearly reached 100%. The rates for secondary education started to increase around 1900, with a more pronounced rise beginning in 1950. India's national five-year plans during the period that followed 1950s placed significant emphasis on female education, with a focus on universalizing elementary education, followed by the push for secondary education (Sarma, 1958). By 2021, both genders exhibited an upward trend in secondary education enrolment. Similarly, the enrolments in tertiary education showed minimal activity until approximately 1950. From 1950 onwards, a steady increase in rates is observed, with boys' enrolment increasing at a faster pace initially. By the year 2000, both boys and girls had significantly higher rates in tertiary education, with boys slightly ahead. However, by 2021, girls' participation in tertiary education had surpassed that of boys. The enrolment of girls in schools has indeed seen remarkable improvements. According to the ASER (2023), the percentage of children aged 6 to 14 enrolled in government schools in rural areas increased from 65.6% in 2018 to 72.9% in 2022. The enrolment rate for girls aged 11 to 14 reached 98% in 2022, a significant advancement compared to the past two decades. Moreover, data from the UDISE+ Report (Ministry of Education, 2022), show that the Gender Parity Index (GPI) stands at 1.02 for elementary and at 1.00 for secondary and 1.1 in tertiary education, indicating that more girls are now enrolling in higher education than boys. These improvements are a testament to the

³¹ A lakh is a unit in the Indian numbering system equal to one hundred thousand (100,000)

³² 15 lakhs = $15 \times 100,000 = 1,500,000$

³³ 96 lakhs = $96 \times 100,000 = 9,600,000$

³⁴ A crore denotes ten million (10,000,000) and is equal to 100 lakhs in the Indian numbering system.

³⁵ 26 crores = $26 \times 10,000,000 = 260,000,000$

effectiveness of targeted government programs and societal shifts in recognizing the importance of girls' education.

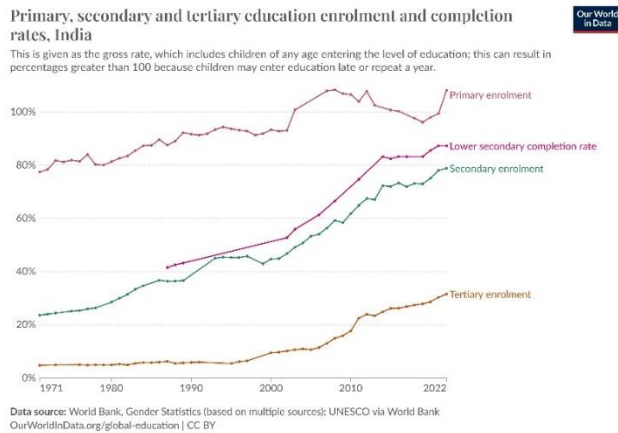


Figure 44. Enrolment and Completion Rates for Primary, Secondary, and Tertiary Education in India (1971-2022)

Source: *Our World in Data* (2024y).

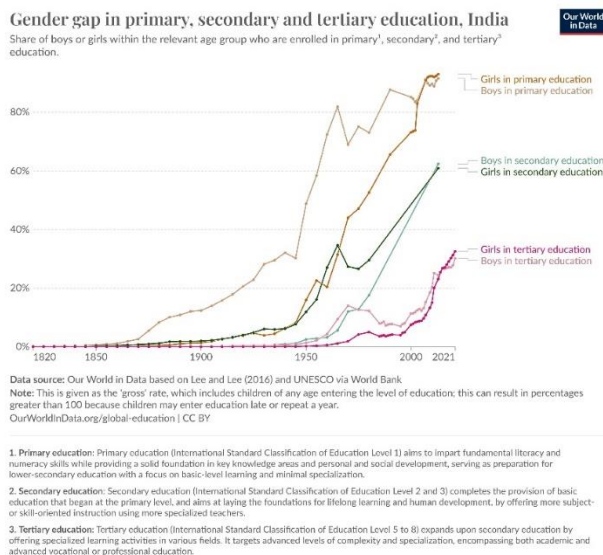


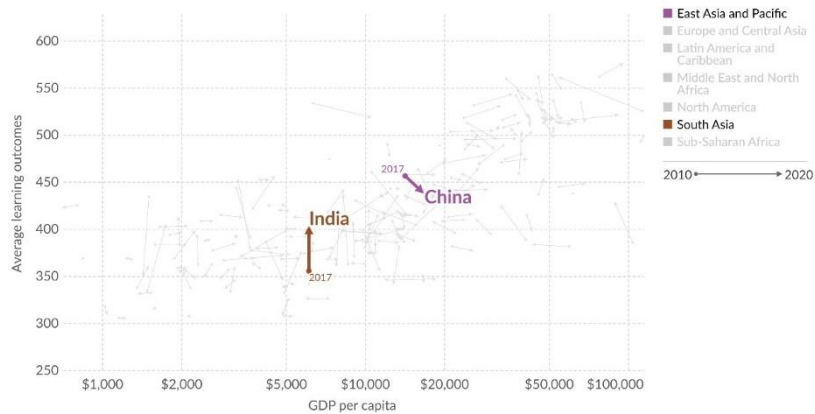
Figure 45. Gender Gap in Primary, Secondary, and Tertiary Education in India (1820-2021)

Source: *Our World in Data* (2024r).

Average learning outcomes vs. GDP per capita, 2010 to 2020

Average learning outcomes correspond to harmonized¹ scores across standardized, psychometrically-robust international and regional student achievement tests.

Our World
in Data



Data source: Patrinos and Angrist (2018) and UNESCO via World Bank; Data compiled from multiple sources by World Bank

Note: GDP per capita data is expressed in international-\$² at 2017 prices.

OurWorldInData.org/global-education | CC BY

1. **Harmonized test scores:** Harmonized test scores consolidate data from several international student achievement testing programs, enabling a standardized comparison of educational attainment across different educational systems and cultures. These scores are measured in TIMSS (Trends in International Mathematics and Science Study) - equivalent units, with 300 denoting minimal attainment and 625 representing advanced attainment.

2. **International dollars:** International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: [What are Purchasing Power Parity adjustments and why do we need them?](#)

Figure 46. Average Learning Outcomes vs. GDP per Capita in India and China (2010-2020)

Source: *Our World in Data* (2024g).

Figure 46 shows that India displays lower GDP per capita and lower average learning outcomes compared to China. Thus, despite economic growth, India's learning outcomes are relatively modest, indicating potential issues in the educational system that are not solely tied to economic factors. China, on the other hand, exhibits significantly higher learning outcomes for its GDP per capita, suggesting more efficient allocation of economic resources in enhancing educational performance.

Similar to China, India has undergone significant demographic changes over the past decades. The use of contraceptives among Indian women has seen a marked increase since 1971 (Figure 47), indicative of improved access to and acceptance of family planning methods, improved reproductive health and autonomy for women. Studies have shown that access to contraception allows women to plan their families and invest more

in their education and careers (Bearak et al., 2021). While the fertility rate in India has significantly decreased from around five births per woman in 1950 to about two births per woman in 2021 (Figure 49), the average age at marriage in India has gradually increased from around 19 years in 1992 to over 21 years in 2015 (Figure 48). Older age at marriage is associated with better maternal and child health outcomes, as women who marry later are more likely to have planned pregnancies and access healthcare.

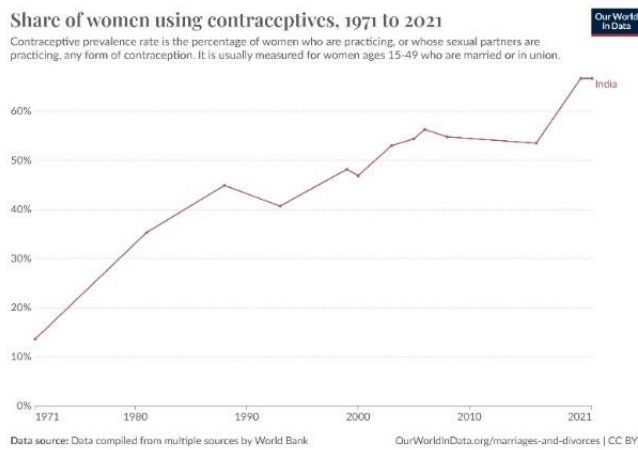


Figure 47. Share of Women Using Contraceptives in India (1971-2021).

Source: *Our World in Data* (2024ae).

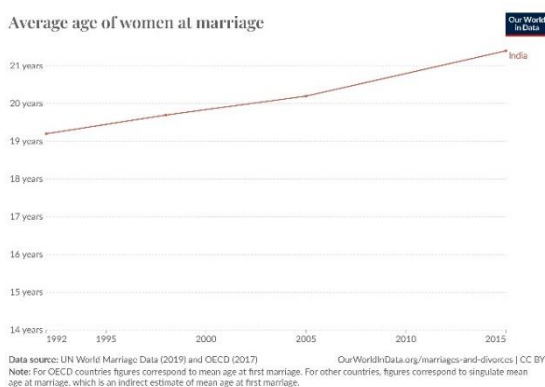


Figure 48. Average Age of Women at Marriage in India (1992-2015)

Source: *Our World in Data* (2024f).

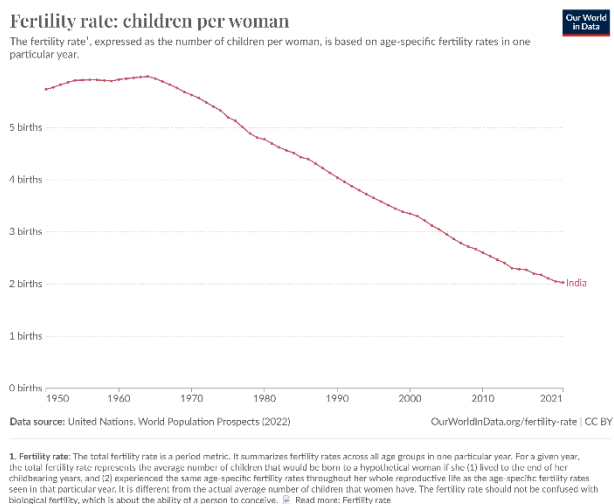


Figure 49. Fertility Rate in India (1950-2021).

Source: *Our World in Data* (2024o).

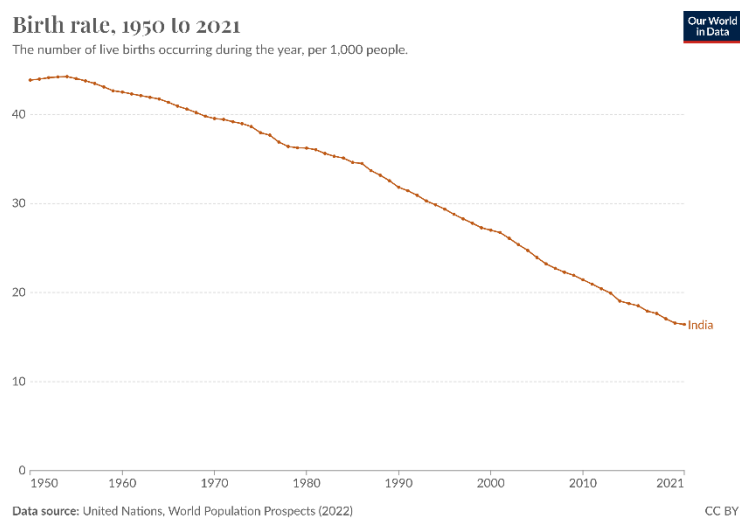


Figure 50. Birth Rate in India (1950-2021).

Source: *Our World in Data* (2024w)

Sex ratio at birth, 1950 to 2021

The sex ratio at birth is measured as the number of newborn boys for every 100 newborn girls. Higher values indicate a much higher number of newborn boys than girls.

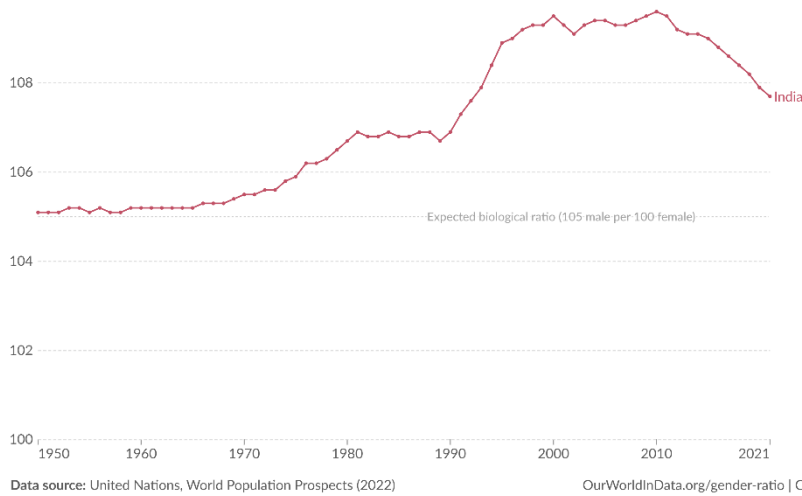


Figure 51. Sex Ratio at Birth in India (1950-2021).

Source: *Our World in Data* (2024ad).

It should be noted that the sex ratio at birth in India, as in China, has been a point of concern, as it deviates from the expected biological ratio (Bongaarts and Guilmoto, 2015). In recent years, there has been a decline in the male-to-female ratio at birth, suggesting improvements but still indicating a preference for male children (Figure 51). According to Meenakshi and Bakshi (2022), the National SRB Index improved from 918 in 2014-15 to 934 in 2019-2020: 422 out of 640 districts covered under BBBP showed improvement in SRB within these intervals. In India, women are the social group most evidently afflicted by discrimination, biases and violence, with prevalent issues such as sexual violence, domestic abuse, and dowry-related deaths (Rani and Verma, 2022). At the same time, higher levels of education and employment among women are commonly associated with higher contraceptive use, allowing women to delay childbirth and pursue higher education and career opportunities, leading to higher FLFP rates. However, societal norms and economic conditions still present significant barriers for these results to be applied in the Indian context. Despite improvements in contraceptive use, delayed marriage, and lower fertility rates, the FLFP rate has seen a slight decline from the mid-2000s (Figure 52), with minority groups and ethnic discrimination also persisting. The

next section aims to explore the effects of these trends on employment dynamics and the participation of women in the workforce.

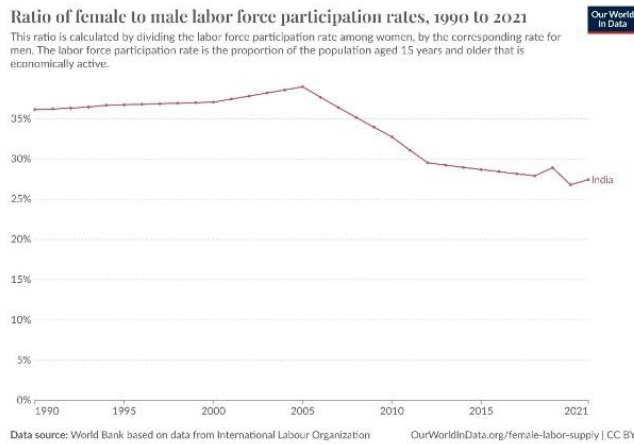


Figure 52. Ratio of Female to Male Labour Force Participation Rates in India (1990-2021).

Source: *Our World in Data* (2024ac).

Employment Dynamics and Gender Inequality in Indian Economic Growth

India presents an intriguing case study for examining the dynamics between female labour participation and educational attainment. Unlike China, it has often been cited for exhibiting a U-shaped relationship between FLFP and educational attainment (Chattopadhyay and Chowdhury, 2022). This pattern suggests that as women reach higher levels of education, their participation in the labour market initially declines but increases significantly at higher educational levels.

India has been one of the fastest-growing major economies in recent decades. Post-liberalization, between 1991 and 2019, GDP growth averaged about 6-7% annually (The World Bank, 2022). The COVID-19 pandemic caused a contraction in 2020, but the economy has been on a recovery path since 2021, with a growth rate of 7.2% in 2022 (ibid.). India's GDP per capita has increased substantially as well, though it remains lower

than that of China. In 1991, India's GDP per capita was approximately \$367³⁶, but by 2020, it had risen to around \$2,000 (World Bank, 2020), reflecting significant but uneven economic development. India has also attracted large FDI, receiving in 2020 FDI inflows of \$64 billion, making it one of the top destinations for foreign investment globally (UNCTAD, 2021). India's labour productivity growth has been slower compared to China, partly due to the larger share of the workforce engaged in low-productivity agricultural activities. Nonetheless, it has improved in recent years, driven by technological adoption and structural reforms.

India saw a significant increase in employment between 1975 and 1984. The nation's public sector expanded significantly during this time. Many jobs were following the government's massive investments in infrastructure projects and nationalisation of important sectors. In addition, the Green Revolution, which started in the late 1960s and lasted into the 1980s, generated jobs in rural areas and changed agricultural production. In pursuit of a state-led industrialization policy, the Indian government nationalised banks, steel, and coal, generating demand for goods and services, which in turn stimulated job development in adjacent industries in addition to directly creating jobs in these sectors (Mukherji, 2009). After 1984, however, employment growth slowed considerably, mostly as a result of public sector layoffs and a lack of job creation in the organised private sector. Employment growth has decreased significantly between 1984 and the present as compared to the 1975-1984 period (Mitra, 2021). COVID-19 had a devastating effect on the Indian economy, making the country's already serious job problems worse (ibid.), with post-COVID growth being comparatively jobless relative to the rising workforce. Large numbers of jobs were lost after the lockdowns, especially in the informal sector, which employs the majority of people and women in India (Mohapatra, 2012). This caused the demand to decline significantly, which in turn caused the economy to contract.

India's labour market participation shows notable differences between rural and urban areas as well as genders (Deshpande and Singh, 2021). Poverty-related considerations frequently drive women's employment in rural areas, where a large number work in agriculture (ibid.). Furthermore, the number of children per household and its size, the accessibility of health and educational resources, all have an impact on

³⁶ Current USD.

how many women enter the workforce (Klasen et al., 2020). In cities, road infrastructure and literacy rates have an impact on women's participation in the labour market, in addition to inadequate health and demographic pressures (Mitra, 2021). Yet, the rise of the service sector, industrialization, and urbanisation all have a favourable impact on the female engagement. Bridging the labour supply and demand gap can be achieved, in part, by enhancing infrastructure, health, and education (Deshpande and Singh, 2021). Therefore, the government's ongoing challenge is to create jobs to support India's expanding young population (Bertelsmann Stiftung, 2024a).

India's FLFP rate is extremely low and is currently falling. Numerous issues, such as social norms, a lack of employment possibilities that align with women's domestic obligations, and unstable economic conditions are responsible for this fall. The volatility of FLFP is typified by frequent changes in status from being in the labour force to being out of the labour force, which makes measures addressing these dynamics necessary to maintain stability and promote long-term involvement ones (Deshpande and Singh, 2021). While women with middle-class education levels are less likely to participate in the labour force, individuals with very low or very high education levels are more likely to do so. Women who have completed higher education, for example, have a greater labour participation rate; in 1999, their positive marginal effect was 21% points; however, by 2011, this had decreased to 14 percentage points (Klasen et al., 2020). Although a higher education boosts employability, the U-shaped pattern indicates that lower participation rates may result from intermediate education levels not offering enough work prospects, due to the prevalence of informal and unstable labour marketplaces (Mohapatra, 2012). Thus, women's alternatives are limited by a lack of diversity in the workforce, which typically confines them to family businesses and agricultural work (Deshpande and Singh, 2021). Moreover, it is crucial to note that FLFP figures risk not to be accurate due to women's sporadic and transient participation in the labour force. One of the biggest obstacles to women entering the labour force is still the difference in pay between genders (Klasen et al., 2020). In addition to impairing women's financial capacity, salary differences, despite equal training and experience, also reduce their willingness to enter the workforce. To create a fairer labour market, broad policy initiatives are necessary to address these gaps. Besides, the disproportionate share of unpaid caregiving duties severely restricts females' career options (Hooda, 2021). To

increase job prospects, it is essential to have a more refined understanding of the factors influencing the demand for women's labour and to implement effective policy responses.

Chapter III: While we cannot alter the direction of the wind of change, we can adjust the sails. A Comparative Analysis

The Role of Institutional Frameworks

The institutional frameworks of China and India have profoundly influenced their approaches to gender equality, sustainable development, and educational initiatives. India's institutional structure, characterized by a complex bureaucracy inherited from British colonial rule and a democratic political system, divides authority between federal and state governments, each with jurisdiction over specific issues (Fleischman, 2015; Rao and Singh, 2006). Policy implementation is managed by the bureaucracy under the oversight of the Union Public Service Commission (Ghosh, 2018). Despite judicial independence, the Indian judiciary is frequently criticized for being overworked and understaffed, resulting in delays in the administration of justice (Bertelsmann Stiftung, 2024a). India also features a free press and a vibrant civil society, though media freedom faces challenges due to political pressures. Conversely, China's institutional framework is defined by a single-party system under the CPC, which centralizes authority over governance and policy. All levels of government are controlled by the CPC, emphasizing centralized decision-making. While local governments are responsible for policy implementation, the central government retains significant control over policy direction. Civil society, media, and online speech are heavily regulated by the state, and the judiciary lacks impartiality, with court decisions often influenced by the CPC (Bertelsmann Stiftung, 2024b).

Globalization and liberalization have driven India's economic growth, particularly in the service sector. However, inadequate infrastructure poses challenges, making it crucial to maintain investment levels to attract foreign capital. Corruption and excessive bureaucracy discourage investments and impede project execution. Economic performance varies significantly among states, with Maharashtra and Gujarat outperforming states like Bihar and Uttar Pradesh (Bertelsmann Stiftung, 2024a). In contrast, China's centralized planning has facilitated the execution of large-scale infrastructure projects, promoting industrialization and urbanization (Bertelsmann Stiftung, 2024b). There are notable geographical and regional disparities, with coastal

areas being significantly more developed than inland regions, analogous to India's rural-urban divide.

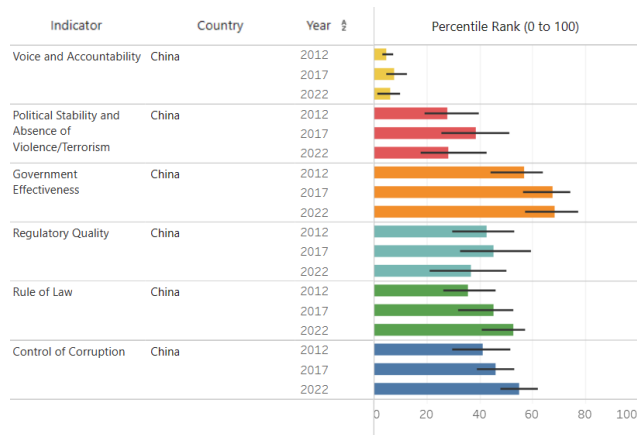


Figure 53. Worldwide Governance Indicators for China (2012-2022).

Source: Kaufmann and Kraay, 2023.

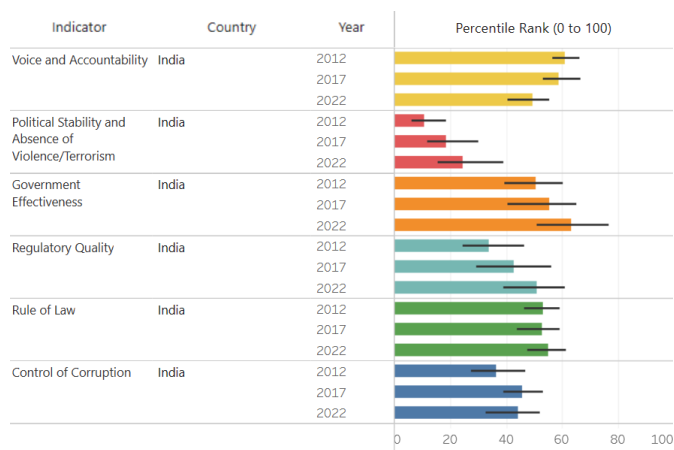


Figure 54. Worldwide Governance Indicators for India (2012-2022).

Source: Kaufmann and Kraay, 2023.

The World Governance Indicators (Kaufmann and Kraay, 2023) provide a comprehensive assessment of institutional and governance aspects such as regulatory quality, rule of law, and corruption in China and India. Figures 53 and 54 illustrate perceptions of public services, civil service capacity, resistance to political pressure, policy quality, and adherence to government programs. The capacity of the civil service, the quality of public services, and the degree of political pressure resistance are all components of government effectiveness. The latter also includes the quality of policy formulation and implementation and the government's commitment to such policies. India has shown improvements despite significant challenges, including an inefficient bureaucracy, inadequate governmental services, and difficulties in policy execution. The weakening of democratic institutions under the BJP-led government has further impeded potential for sustainable growth (Bertelsmann Stiftung, 2024a). China, while performing better in some areas, faces issues such as political interference, corruption, and infrastructural deficiencies that limit government efficacy. The centralized control system in China facilitates efficient policy implementation and public service delivery, though corruption and lack of transparency occasionally undermine effectiveness (Bertelsmann Stiftung, 2024b).

India's regulatory quality is hindered by inconsistent frameworks, burdensome guidelines, and generally poor standards, exacerbated by frequent policy changes and bureaucratic red tape. In contrast, China's regulatory structures are more effective, with well-defined policies and enforcement procedures, although overbearing government regulations and restrictions present challenges. The Indian legal system, while weaker in some respects than China's, upholds fundamental legal principles and protects rights despite delays, judicial corruption, and lax law enforcement. In China, the rule of law faces significant obstacles due to CPC influence over the judiciary, leading to arbitrary law enforcement and a lack of judicial independence. This is compounded by issues of transparency and human rights abuses (Bertelsmann Stiftung, 2024b). Both countries have experienced poverty reduction due to economic progress. However, social inequality has increased in both contexts despite their differing institutional frameworks.

Sustainable Development Goals: A Snapshot of Progress and Challenges

This section provides an analysis of the Sustainable Development Goals (SDGs) dashboards for China and India. Adopted by the United Nations in 2015, the SDGs encompass 17 goals aimed at achieving a sustainable and equitable world by 2030 (United Nations, 2024). The dashboards highlight each country's progress, areas of achievement, and areas needing further attention. China is ranked 63rd globally with a score of 72.01, while India is 112th with a score of 63.45 (United Nations, 2023).



Figure 55. China's SDG Dashboard 2023.

Source: *Sustainable Development Report 2023* (United Nations, 2023).



Figure 56. India's SDG Dashboard 2023.

Source: Sustainable Development Report 2023 (United Nations, 2023).

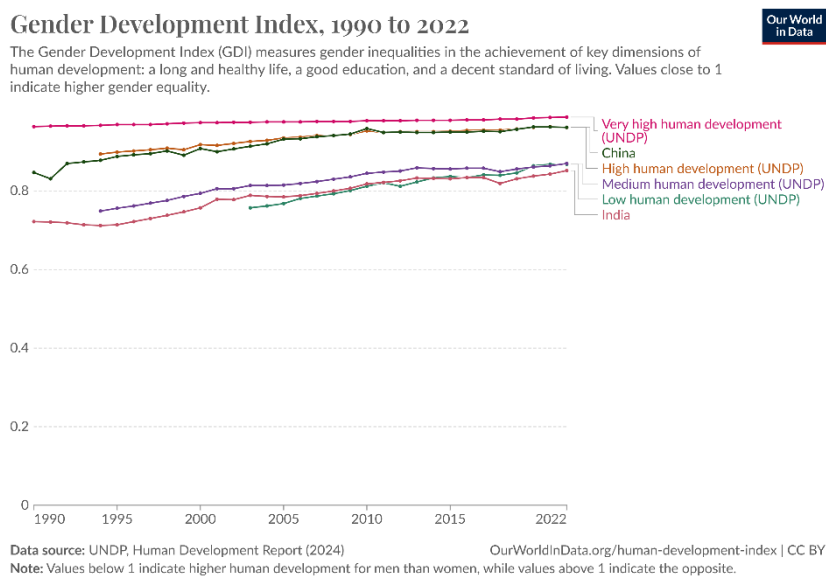


Figure 57. Gender Development Index (1990-2022) Comparing Very High, High, Medium, and Low Human Development with China's and India's Levels

Our World in Data (2024q)

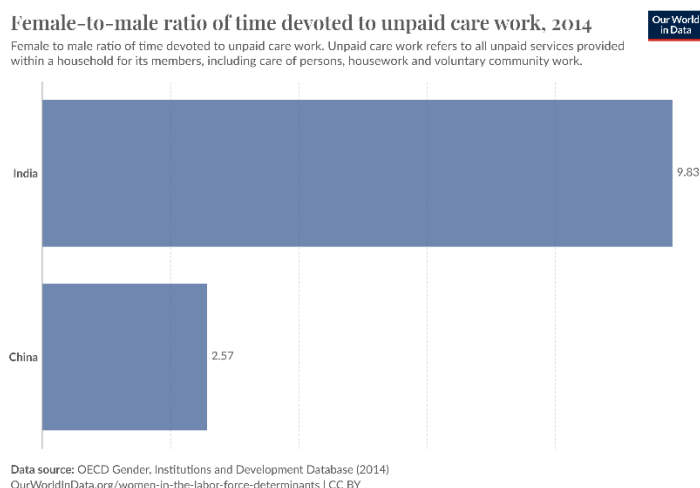


Figure 58. Female-to-Male Ratio of Time Devoted to Unpaid Care Work in India compared to China.

Source: *Our World in Data* (2014).

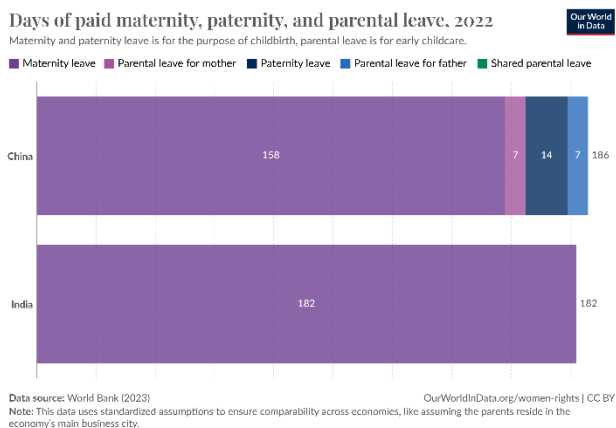
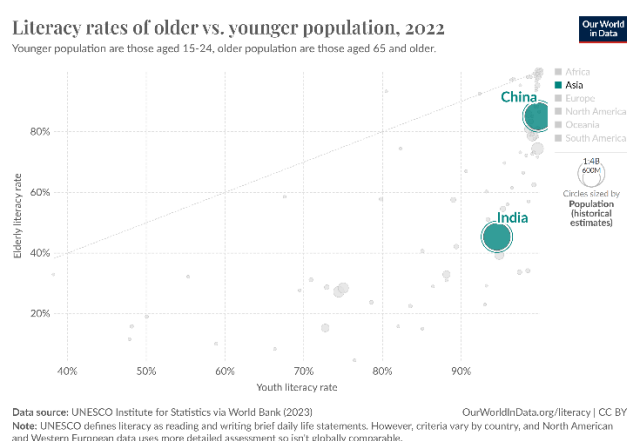


Figure 59. Days of Paid Paternity, Maternity, and Parental Leave in China and India (2022).

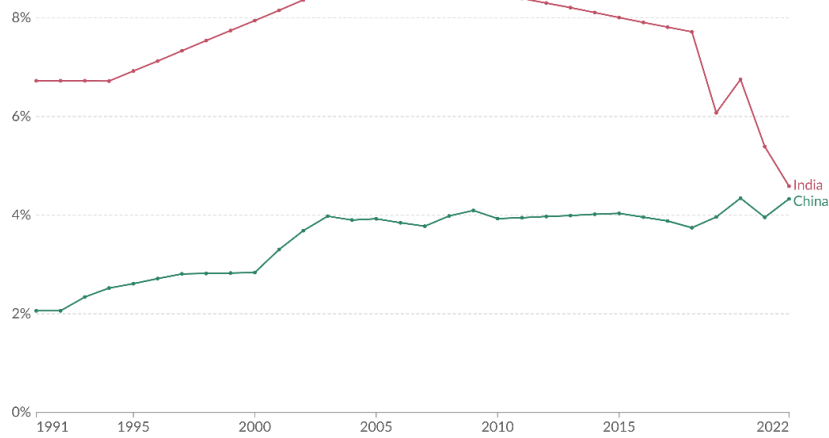
Source: *Our World in Data* (2024j)

Gender equality remains a significant challenge for China and a major challenge for India, as evidenced by their SDG dashboards (Figures 55 and 56). China's relative improvement in the GDI indicates a more effective approach to gender equality. In contrast, India's slower progress highlights the need for targeted policy interventions to address deep-seated gender biases and socio-economic barriers. The high female-to-male ratio in unpaid care work in India reflects traditional gender roles and a lack of support systems (OECD, 2014), whereas China's lower ratio suggests more equality in domestic responsibilities (Figure 58). Similar implications can be drawn from Figure 59, where China offers 158 days of maternity leave, 7 days of paternity leave, and 14 days of parental leave, while India provides 182 days of maternity leave but no paternity or parental leave. An analogous trend is shown for Quality Education (SDG 4), with China's status on track, while India faces significant challenges but is showing moderate improvement. This is reflected in literacy rates and public spending on education.



Unemployment rate, women, 1991 to 2022

The unemployment rate expresses the number of people unemployed as a percent of the labor force. All figures correspond to estimates modeled by the source.



Data source: Multiple sources compiled by World Bank (2024)

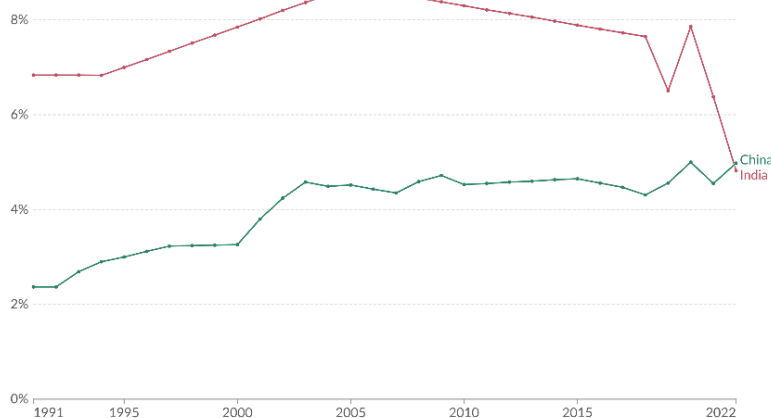
OurWorldInData.org/female-labor-supply | CC BY

Figure 62. Women Unemployment Rate Comparison Between China and India (1991-2022).

Source: *Our World in Data* (2024ag).

Unemployment rate, 1991 to 2022

Unemployment refers to the share of the labor force that is without work but available for and seeking employment.



Data source: Multiple sources compiled by World Bank (2024)

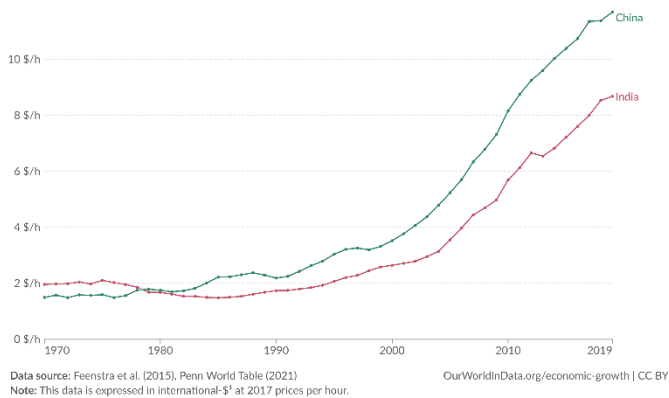
OurWorldInData.org/global-education | CC BY

Figure 63. Unemployment Rate Comparison Between China and India (1991-2022).

Source: *Our World in Data* (2024af).

Productivity: output per hour worked

Productivity is measured as gross domestic product (GDP) per hour of work. This data is adjusted for inflation and differences in the cost of living between countries.



1. International dollars: International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: What are Purchasing Power Parity adjustments and why do we need them?

Figure 64. Chinese and Indian Productivity in Comparison (1970-2019).

Source: *Our World in Data* (2024z)

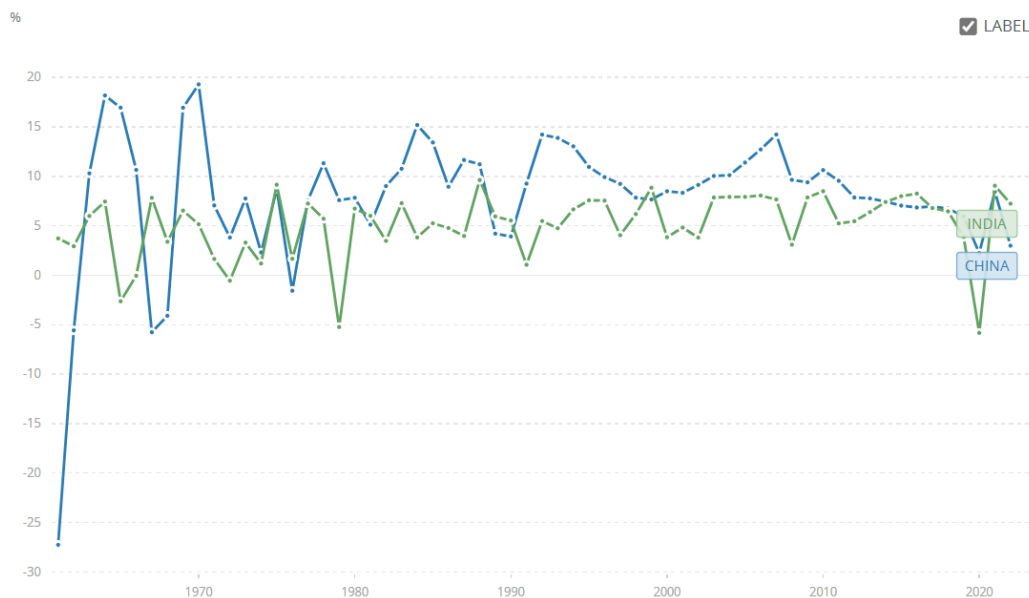


Figure 65. Annual Percentage of GDP Growth in India and China.

Source: *World Bank national accounts data and OECD National Accounts data files* (2022).

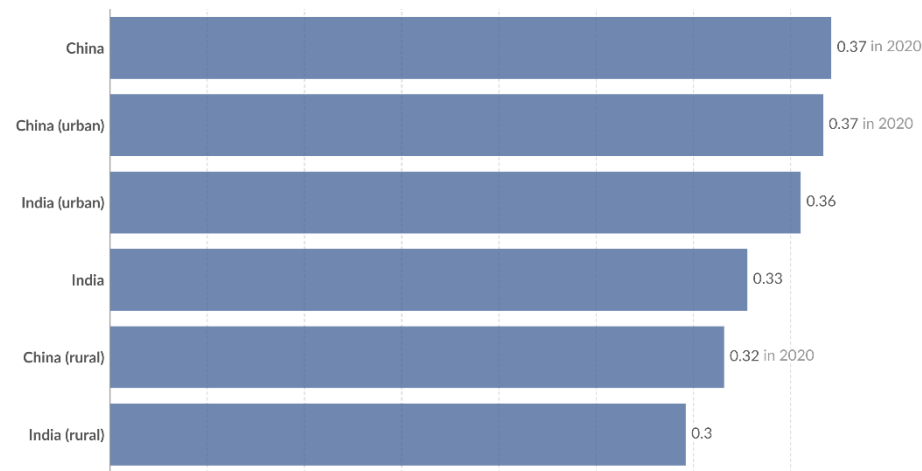
The SDG Dashboard displays how both countries face challenges in achieving SDG 8, but progress is being made. China's productivity has significantly increased since

the 1990s due to industrialization and economic reforms, whereas India's growth has been slower, reflecting ongoing infrastructure and workforce development challenges (Figure 64). Female unemployment rates show different trends in China and India. China's rate has been rising, possibly due to economic restructuring, while India's rate has been declining, indicating slight improvements in female labour force participation (Figures 62 and 63). In China, positive state interventions promoting gender equality, accessibility of childcare, and the necessity of dual-wage support for families contribute to higher female employment rates. However, employers still practice discrimination, and trade unions or women's associations tend to focus on welfare rather than bargaining (Sharma and Sharma, 2019). In contrast, Indian women experience lower participation rates, with a significant concentration in the unorganized sector and informal employment. High unemployment rates among women with low skills and literacy, coupled with a substantial gender pay gap, exacerbate the issue. Cultural factors such as caste, religion, and male-dominant values that view women as inferior further hinder women's employment opportunities (Borooah and Iyer, 2005). Discriminative practices by employers and the disinterest of trade unions in the informal sector, where many women are employed, add to the challenges (Sharma and Sharma, 2019). In India, only 29.7% of females with college degrees in rural areas and 27.9% in urban areas participate in the workforce (Mukherjee, 2015). For those with professional diplomas and certificates, the figures are slightly higher but still reflect a significant gap (*ibid.*). Similarly, in China, female unemployment rates increased between 2001 and 2008, even as male unemployment declined. Urban China experienced higher unemployment rates for females across various age cohorts, mirroring the situation in India where female unemployment rates are nearly double those of males (*ibid.*). Therefore, the gender labour force gap is evident, and closing it in emerging markets and developing economies by just 5.9 percentage points is expected to increase real GDP by about 8 percentage points (Sayeh, Badel, and Goyal, 2023).

Income inequality: Gini coefficient, 2021



The Gini coefficient¹ measures inequality on a scale from 0 to 1. Higher values indicate higher inequality. Depending on the country and year, the data relates to income measured after taxes and benefits, or to consumption, per capita².



Data source: World Bank Poverty and Inequality Platform (2024)

OurWorldInData.org/economic-inequality | CC BY

Note: Income and consumption estimates are available separately in this [Data Explorer](#).

1. **Gini coefficient:** The Gini coefficient is the most commonly used measure of inequality. It is typically used as a measure of income inequality, but it can be used to measure the inequality of any distribution – such as the distribution of wealth, or even life expectancy. It measures inequality on a scale from 0 to 1, where higher values indicate higher inequality. This can sometimes be shown as a percentage from 0 to 100%, this is then called the 'Gini Index'. A value of 0 indicates perfect equality – where everyone has the same income. A value of 1 indicates perfect inequality – where one person receives all the income, and everyone else receives nothing. Read more in our article: [Measuring inequality: What is the Gini coefficient?](#)

2. **Per capita:** 'Per capita' here means that each person (including children) is attributed an equal share of the total income received by all members of their household.

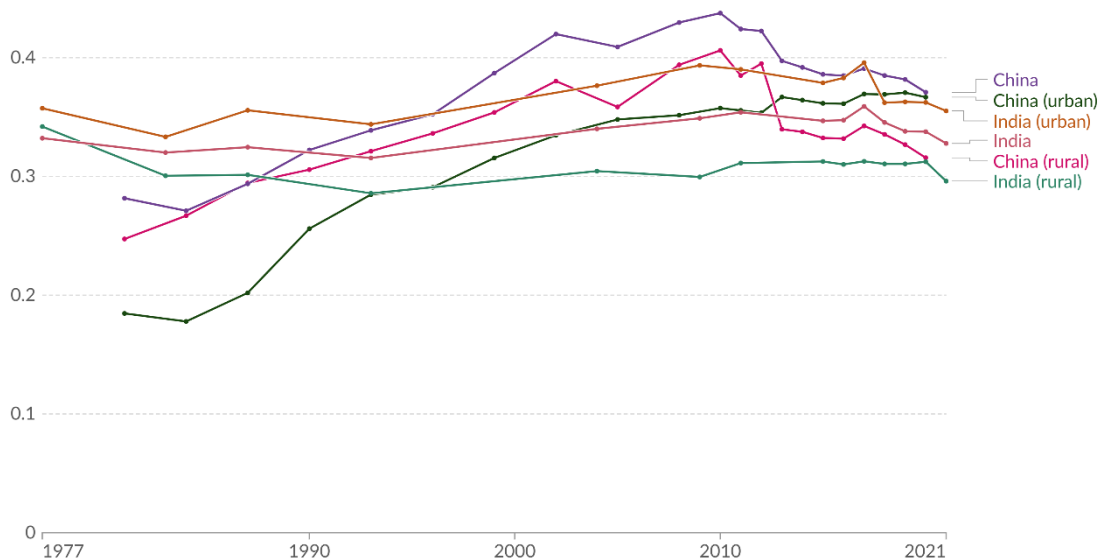
Figure 66. Income Inequality Across the Rural-Urban Divide in China and India (2021)

Source: *Our World in Data* (2024k)

Income inequality: Gini coefficient, 1977 to 2021

Our World
in Data

The Gini coefficient¹ measures inequality on a scale from 0 to 1. Higher values indicate higher inequality. Depending on the country and year, the data relates to income measured after taxes and benefits, or to consumption, per capita².



Data source: World Bank Poverty and Inequality Platform (2024)

OurWorldInData.org/economic-inequality | CC BY

Note: Income and consumption estimates are available separately in this [Data Explorer](#).

1. **Gini coefficient:** The Gini coefficient is the most commonly used measure of inequality. It is typically used as a measure of income inequality, but it can be used to measure the inequality of any distribution – such as the distribution of wealth, or even life expectancy. It measures inequality on a scale from 0 to 1, where higher values indicate higher inequality. This can sometimes be shown as a percentage from 0 to 100%, this is then called the 'Gini Index'. A value of 0 indicates perfect equality – where everyone has the same income. A value of 1 indicates perfect inequality – where one person receives all the income, and everyone else receives nothing. Read more in our article: [Measuring inequality: What is the Gini coefficient?](#)

2. **Per capita:** 'Per capita' here means that each person (including children) is attributed an equal share of the total income received by all members of their household.

Figure 67. Income Inequality Across the Rural-Urban Divide in China and India Over the Years (1977-2021).

Source: *Our World in Data* (2024k)

Analysing the increase in China's Gini coefficient from its baseline is particularly important for this study. Figure 67 highlights the challenges of managing income inequality in the course of rapid economic growth and the impact of the rural-urban divide, further complicated by the hukou system. Analogously, India faces major challenges in addressing caste-based inequalities and a similar rural-urban divide, contributing to its own income disparity issues. Their progress on SDG 10 indicates that while China's status appears to be stagnating in terms of income inequality, India's status

is deteriorating, showing a serious complication in the path towards sustainable growth. This research demonstrates that rapid economic development does not always improve unobserved qualifications or reduce discrimination. Thus, a key policy priority should be to empower women in society and change how women are perceived in the labour market to create a more inclusive and fairer environment. The outcomes mirroring the SDGs and the FLFP patterns underscore the importance of integrating more behavioural economics principles into policy design to achieve sustainable social change.

Why Do China and India Need Behavioural Policies and Nudging?

The gender labour force participation gap is a persistent issue that economic development alone has not resolved. It might narrow, but it is unlikely to fully close for many countries, leaving women unable to fully utilize their abilities and talents to the detriment of society (Badel and Goyal, 2024). Such gap is an issue of supply and demand: through the adoption of a behavioural perspective, it becomes possible to add nuance to this macroeconomic issue and consider psychological factors and framing. Current policy efforts have not been sufficient to decisively close such gap, and setbacks remain likely due to economic shocks, crises, or policy reversals: more inclusive efforts are required to tackle barriers preventing gender equality in the job market (Badel, Fabrizio, and Goyal, 2024).

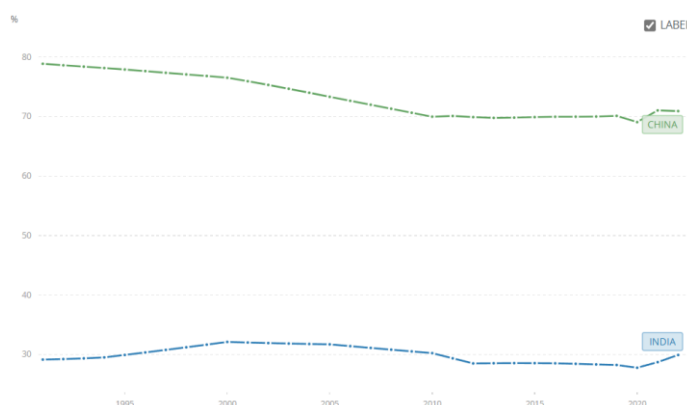


Figure 68. Labour Force Participation Rate Modelled ILO Estimate as A Percentage of The Female Population Aged 15-64.

Source: ILO, 2024.

This study shows that improvements in the educational systems have not proportionally translated into increased FLFP. The trends identified in preceding sections reveal inefficiencies in resource allocation and suggest that conventional policy-making techniques have diminished marginal utility. While educational progress has fostered greater gender equality in access to education, investments in education have yielded diminishing returns in FLFP, necessitating innovative approaches. This research proposes the introduction of behavioural policies grounded in behavioural economics to address these challenges effectively. Central to this approach is the establishment of local nudge units, tailored to the specific needs and conditions of different regions. These units are designed to address the disparities between rural and urban areas in both countries, as evidenced by income inequality indices.

Behavioural economics posits that incentives and constraints can reshape biases and promote changes in behaviour. Integrating local nudge units could further enhance traditional policymaking's impacts, addressing cultural prejudices through policies that mitigate bounded rationality and present bias (Sunstein and Thaler, 2003). For example, introducing nudges such as personalized messages to parents about the long-term benefits of girls' higher education and employing gamification techniques (Deterding et al., 2011) like rewarding schools and workplaces for high female attendance rates can be effective and supportive of the previously mentioned policies. These nudging interventions can mitigate the effects of risk perception and decision-making biases on gender parity and workplace access by leveraging Kahneman and Tversky's prospect theory (1979).

In India, the relationship between education and FLFP, interpreted through the lens of behavioural science, reflects the influence of social norms, cognitive biases, and intra-household dynamics. Similarly, despite improvements in female educational attainment, the Chinese labour market faces challenges in achieving equality due to gender discrimination in hiring practices and promotions, workplace bias, inadequate childcare support. While initiatives such as parental leave slowly started to address some of these issues, persistent gender pay gaps and cultural biases continue to impede progress. For example, existing parental leave policies in China could be more effective if nudges were used to encourage their uptake. Employers could send personalized reminders to fathers about the benefits of taking paternity leave, and workplaces could

gamify the process by awarding recognition or small rewards to departments with the highest uptake rates.

Interventions designed to shift employer biases, such as anonymized job applications and gender bias training, are further proposed to foster fairer hiring practices. Additionally, policies incentivizing shared household responsibilities and supporting women in leadership roles need to be included in future behavioural strategies. This is where nudge units will be the most impactful, as they can provide cost-effective interventions by leveraging cognitive biases and heuristics, such as the availability heuristic identified by Tversky and Kahneman (1973). In resource-constrained environments, these units can optimize policy impacts by subtly guiding citizens towards behavioural changes. Given the rural-urban divide in both China and India, local nudge units with context-specific strategies are essential. Successful implementation necessitates balancing efficacy with respect for individual freedoms, ensuring policies do not appear coercive through public consultations and pilot testing. The objective is to influence public thinking and decision-making to enhance compliance with government policy and international standards, thereby reducing social and governmental costs associated with inaction, inefficient allocation of resources and poor compliance. To illustrate how nudging could enhance policy outcomes and goals achievement, it is important to consider the following success stories.

- Ideas 42, a non-profit organization, applied behavioural science to address intimate partner violence by targeting alcohol consumption³⁷. Rickshaw drivers were incentivized with monetary deposits into a savings account to maintain sobriety while on the job. This nudge led to a 30% drop in intimate partner violence (Smith and Surianarain, 2021). The study demonstrated that a combined incentives and BCT approach is effective in a low-resource setting, highlighting the potential for scalable, cost-effective behavioural interventions (Hartmann et al., 2020).
- Zurich Insurance addressed their gender pay gap in the UK by changing the default option for new job vacancies to be open to part-time, job-share, and full-

³⁷ The research was carried out in the lower-income regions of Jaya Nagar in Bengaluru, India.

time employees. This experiment resulted in a 16.4% increase in female applicants and a 19.3% increase at the senior level (Smith and Surianarain, 2021).

- The Choreful app in Norway split household chores more equitably between partners through gamification, shifting social norms around household responsibilities and reducing the burden on women (Huck, 2022).
- The Speed Camera Lottery in Stockholm rewarded drivers who observed the speed limit with the fees paid by those who violated it, dropping the average speed and reducing the accident rate without restricting choice (Mesko, 2018).

Evidently, the field of economics still predominantly operates within a fundamentally gendered framework. The concept of “Homo Economicus,”³⁸ which suggests individuals are entirely rational and self-interested, implicitly excludes women, who are often perceived as influenced by emotions or concerns for others (Plant et al., 2000). Addressing gender biases and the emotional gender divide in workplace and economic thinking is crucial for creating more inclusive, relevant, and effective policies (Bohnet, 2018). While promising, behavioural sciences and nudging are not panaceas; poorly designed policies cannot be salvaged by nudging alone. Policymakers, who are also susceptible to biases, require decentralized and transparent decision-making processes, empowering local nudge units to design and implement contextually appropriate interventions.

³⁸ Homo Economicus, or “Economic Man,” is a theoretical construct in economics that represents an idealized human being who acts rationally and with complete knowledge, making decisions solely to maximize their utility or economic benefit. This model assumes that individuals have consistent and stable preferences, are fully rational and capable of making logical decisions, possess complete information about all available options, and act in their self-interest to achieve the highest possible personal gain. The concept is used to simplify economic models and predict behaviour in markets and other economic settings. It originates from classical economic thought, notably Adam Smith’s idea of the “invisible hand” guiding self-interested individuals to unintentionally benefit society, and John Stuart Mill’s elaboration on rational economic behaviour.

Conclusions and Policy Recommendations

The primary objective of this thesis was to critically examine the impact of education on economic growth and inequalities in China and India. After analysing past and present policies and initiatives, the study aimed to understand how they reshaped educational attainments, labour market dynamics, societal norms, and gender roles. The focus was on identifying the challenges and opportunities in promoting gender equality and sustainable economic development.

In the first chapter, the findings underscore the pivotal role of China's educational policies and initiatives in promoting gender equality. The OCP controlled population growth and altered societal views on gender roles, leading to improved educational investment in daughters. The CEL mandated nine years of compulsory schooling, reducing gender and educational inequalities within households. The Spring Bud Project enhanced educational opportunities for girls from low-income families, lowering dropout rates and financial barriers. Looking forward, the Education Modernization 2035 plan brings hope for bridging the urban-rural educational quality gap, improving access to education and vocational training in rural regions. China's transition to a market economy post-1978 increased FLFP, particularly in urban areas, but challenges like childcare responsibilities and gender pay gaps persist. Despite significant economic growth from 1978 to 2010, recent growth has slowed, with FLFP stabilizing around 61.6% (World Bank, 2020; Our World in Data, 2024m).

Chapter II addresses India's efforts to tackle historical and socioeconomic inequalities in female education. NEP 2020 emphasizes accessibility, equity, and quality, aiming for universal access to early childhood education by 2025. The BBBP scheme combats gender biases and unequal educational opportunities. The SSA has improved girls' enrolment and retention rates. The NIPUN Bharat Scheme focuses on foundational literacy and numeracy for marginalized girls. Despite improvements in literacy rates and school enrolment, gender disparities remain, particularly in rural areas. India's FLFP shows a U-shaped relationship with educational attainment. Women with very low or high education levels are more likely to participate in the labour force. Despite steady GDP growth since the 1990s, FLFP has declined since the early 2000s (The World Bank, 2022; Our World in Data, 2024ac).

Chapter III reveals differing institutional frameworks in China and India and their impact on gender equality and sustainable development. China's centralized governance facilitates efficient policy implementation, while India's federal structure poses challenges. Both countries have made strides in reducing poverty and improving educational outcomes, but gender and regional disparities persist. SDG dashboards show China ranking higher than India, reflecting different localized challenges. Implementing behavioural policies and local nudge units is proposed to influence public behaviours and promote gender equality. Comprehensive reforms addressing workplace discrimination, childcare support, and equitable hiring practices are necessary. Future research should focus on behavioural economics to challenge cultural and gender biases and develop strategies for sustainable female workforce participation. Long-term solutions should target rural and underprivileged regions, emphasizing inclusive growth and broader social and economic benefits.

This study highlighted several barriers, including the urban-rural divide and the lack of proportional increase in female labour force participation despite educational achievements. Comprehensive reforms addressing workplace discrimination, childcare support, and fair hiring practices are necessary. The application of nudge theory could guide behaviours toward gender-neutral decisions. Future research should focus on the role of behavioural and development economics in challenging cultural and gender biases and developing strategies for sustainable female workforce participation. Long-term solutions should target rural and underprivileged regions, calling attention to inclusive growth and broader social and economic benefits, while ensuring that economic developments benefit all segments of society.

Both China and India have achieved significant economic growth and improved their educational systems, yet the benefits have not been evenly distributed, particularly regarding employment and gender equality. Recognizing the human element, people with their values, stories, and unique perspectives, is essential in addressing policies' failures and successes. Effective interventions should focus on habits, social support, and accessibility, designed through nudge units tailored to local, cultural and social contexts. Relying solely on data such as school enrolment, growth rates, and economic shifts can lead to the failure of development efforts, manifesting in societies' inability to manage contradictions, leverage their talent pool, and drive meaningful change.

The thesis's findings underscore the importance of systemic and structural interventions to achieve the SDGs. Policymakers must create new pathways that facilitate and encourage female participation in the workforce, including developing behavioural strategies that make it attractive for employers to hire women and ensuring a seamless process of entering and, more importantly, staying in the labour market beyond the recruitment phase. Leveraging concepts from behavioural economics, such as default options and social norms, can help create environments where gender equality is the norm, rather than the exception. Behavioural change is likely to become one of the most important solutions to humanity's needs, and countries and international organizations must be armed with more and better guidelines to promote it.

However, significant questions remain: are the efforts and policies implemented so far enough to ensure sustainable and equitable development? The response, unfortunately, is no. What more can be done to dismantle the gender biases that pervade our societies? Is it even possible to achieve gender equality within the current socio-economic structures, or is radical transformation needed? How can educational policies translate into real-world changes in gender dynamics and economic participation? Is the world ready to embrace the complexity of these challenges and commit to the game-changing transformations required for a more equitable future? The answer, inspired by Bohnet's work (2018), lies in the collective willingness to change, by design.

List of Acronyms

ANM:	Auxiliary Nurse Midwife
ASER:	Annual Status of Education Report
ASHA:	Accredited Social Health Activist
AWW:	Anganwadi Worker
BBBP:	Beti Bachao Beti Padhao
BJP:	Bharatiya Janata Party
BTI:	Bertelsmann Transformation Index
CEL:	Compulsory Education Law
CPC:	Communist Party of China
CSR:	Child Sex Ratio
ECCE:	Early Childhood Care and Education
FLFP:	Female Labour Force Participation
FLN:	Fundamental Literacy and Numeracy
FDI:	Foreign Direct Investment
GDI:	Gender Development Index
GDP:	Gross Domestic Product
GER:	Gross Enrolment Ratio
GII:	Gender Inequality Index
GPI:	Gender Parity Index
HDI:	Human Development Index
HEIs:	Higher Education Institutions
ICT:	Information and Communication Technology
ILO:	International Labour Organization
IMF:	International Monetary Fund
KGBV:	Kasturba Gandhi Balika Vidyalaya
NCW:	National Commission for Women
NEP:	National Education Policy
NIPUN:	National Initiative for Proficiency in Reading with Understanding and Numeracy
NPEGEL:	National Programme for Education of Girls at Elementary Level
OCP:	One Child Policy
OECD:	Organization for Economic Co-operation and Development
POA:	Programme of Action
RTE:	Right to Education Act
SCs:	Scheduled Castes
SDGs:	Sustainable Development Goals
SEZs:	Special Economic Zones
SRB:	Sex Ratio at Birth

SSA: Sarva Shiksha Abhiyan

STEM: Science, Technology, Engineering, and Mathematics

STs: Scheduled Tribes

UN: United Nations

UNCTAD: United Nations Conference on Trade and Development

UNDP: United Nations Development Programme

UNESCO: United Nations Educational, Scientific and Cultural Organization

WTO: World Trade Organization

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